

Gaussian Filters and “Markov” Localization

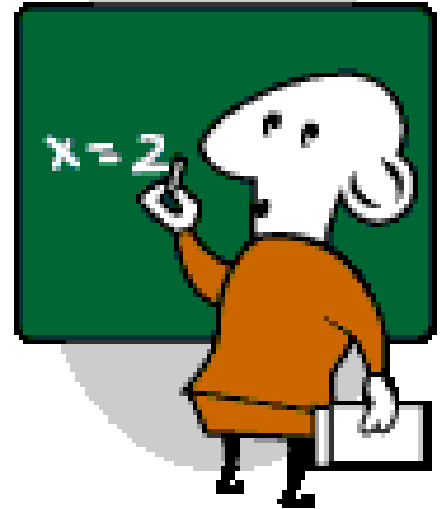
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*Lecture slides heavily use material from the textbook and
Sebastian Thrun, Lecture Slides; <http://www.probabilistic-robotics.org/>*



What we will discuss

- Review the Linear-Gaussian assumption and implications,
- Discuss the Kalman Filter – KF implementation of the Bayes Filter,
- Discuss nonlinear systems and linearization through Taylor Series: Extended Kalman Filter – EKF,
- Discuss alternative linearization: Unscented Kalman Filter – UKF
- Localization through EKF and UKF,
- Multi-Hypothesis Kalman Filter,
- Information Filter,
- Summary





Bayes Filter Reminder

```
1:   Algorithm Bayes_filter( $bel(x_{t-1}), u_t, z_t$ ):  
2:     for all  $x_t$  do  
3:        $\overline{bel}(x_t) = \int p(x_t \mid u_t, x_{t-1}) bel(x_{t-1}) dx_{t-1}$   
4:        $bel(x_t) = \eta p(z_t \mid x_t) \overline{bel}(x_t)$   
5:     endfor  
6:     return  $bel(x_t)$ 
```

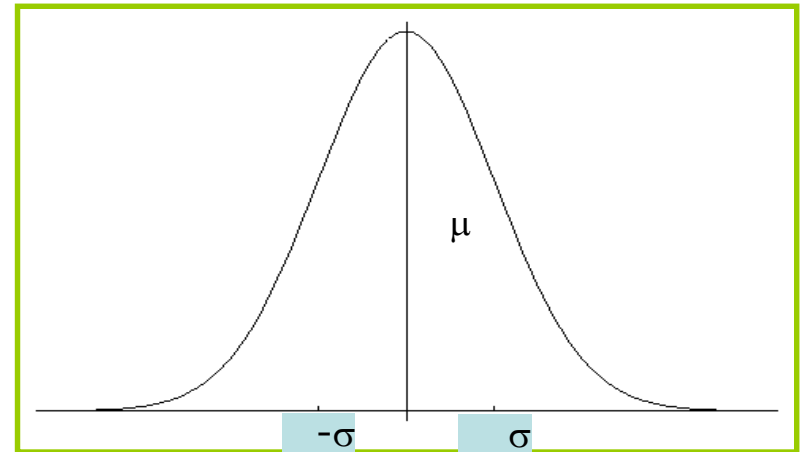


Gaussian pdfs

$$p(x) \sim N(\mu, \sigma^2):$$

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

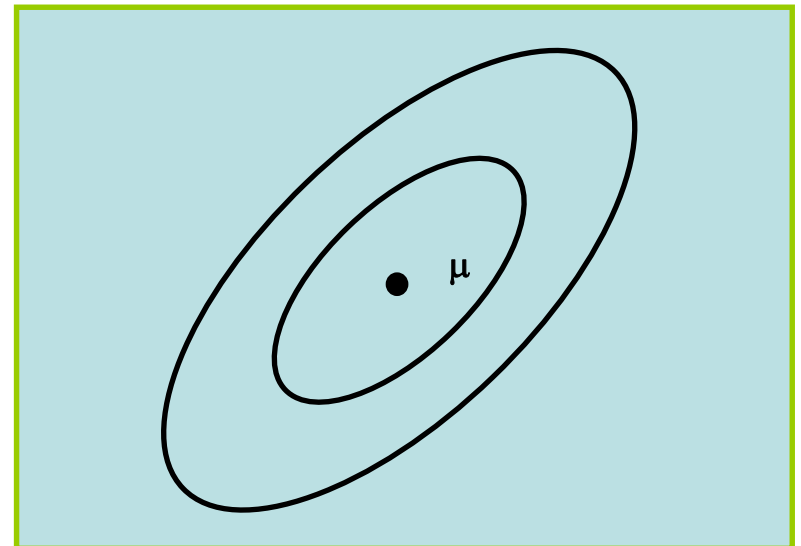
Univariate



$$p(\mathbf{x}) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}):$$

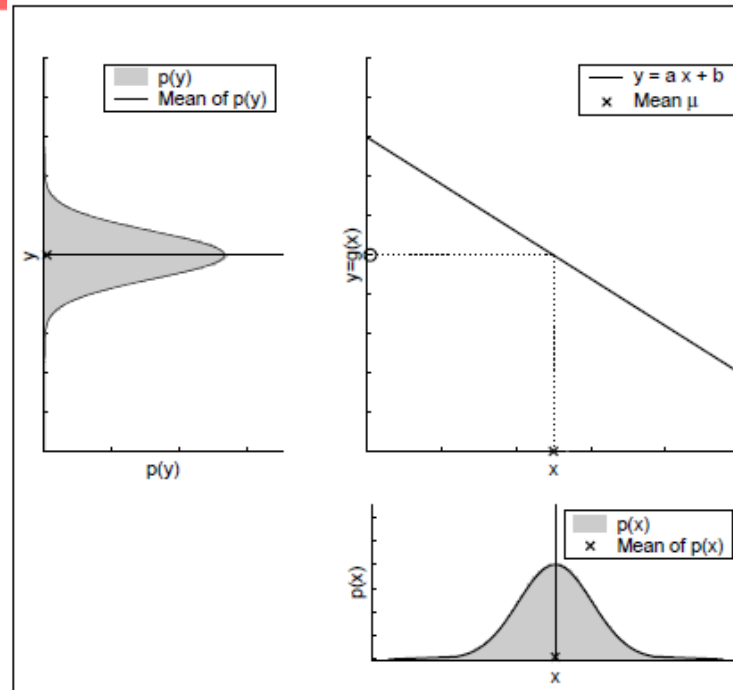
$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} e^{-\frac{1}{2} (\mathbf{x}-\boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu})}$$

Multivariate





Properties of Gaussian pdfs



$$\left. \begin{array}{l} X \sim N(\mu, \sigma^2) \\ Y = aX + b \end{array} \right\} \Rightarrow Y \sim N(a\mu + b, a^2\sigma^2)$$

$$\left. \begin{array}{l} X_1 \sim N(\mu_1, \sigma_1^2) \\ X_2 \sim N(\mu_2, \sigma_2^2) \end{array} \right\} \Rightarrow p(X_1) \cdot p(X_2) \sim N\left(\frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2} \mu_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \mu_2, \frac{1}{\sigma_1^{-2} + \sigma_2^{-2}}\right)$$



Multivariate Gaussian pdfs

$$\left. \begin{array}{l} X \sim N(\mu, \Sigma) \\ Y = AX + B \end{array} \right\} \Rightarrow Y \sim N(A\mu + B, A\Sigma A^T)$$

$$\left. \begin{array}{l} X_1 \sim N(\mu_1, \Sigma_1) \\ X_2 \sim N(\mu_2, \Sigma_2) \end{array} \right\} \Rightarrow p(X_1) \cdot p(X_2) \sim N\left(\frac{\Sigma_2}{\Sigma_1 + \Sigma_2} \mu_1 + \frac{\Sigma_1}{\Sigma_1 + \Sigma_2} \mu_2, \frac{1}{\Sigma_1^{-1} + \Sigma_2^{-1}}\right)$$

- We stay in the “Gaussian world” as long as we start with Gaussians and perform only linear transformations.



The Discrete Kalman Filter

Estimates the state $\mathbf{x}$ of a discrete-time controlled process that is governed by the linear stochastic difference equation

$$x_t = A_t x_{t-1} + B_t u_t + \varepsilon_t$$

with a measurement

$$z_t = C_t x_t + \delta_t$$

Where all random variables have Gaussian pdfs:
Called “*Linear-Gaussian System*”



Linear-Gaussian System

$$x_t = A_t x_{t-1} + B_t u_t + \varepsilon_t$$

$$z_t = C_t x_t + \delta_t$$

 A_t

Matrix ($n \times n$) that describes how the state evolves from $t-1$ to t without controls or noise.

 B_t

Matrix ($n \times l$) that describes how the control u_t changes the state from $t-1$ to t .

 C_t

Matrix ($k \times n$) that describes how to map the state x_t to an observation z_t .

 ε_t

Random variables representing the process and measurement noise that are assumed to be white, independent and Gaussian distributed with covariance R_t and Q_t respectively.

 δ_t

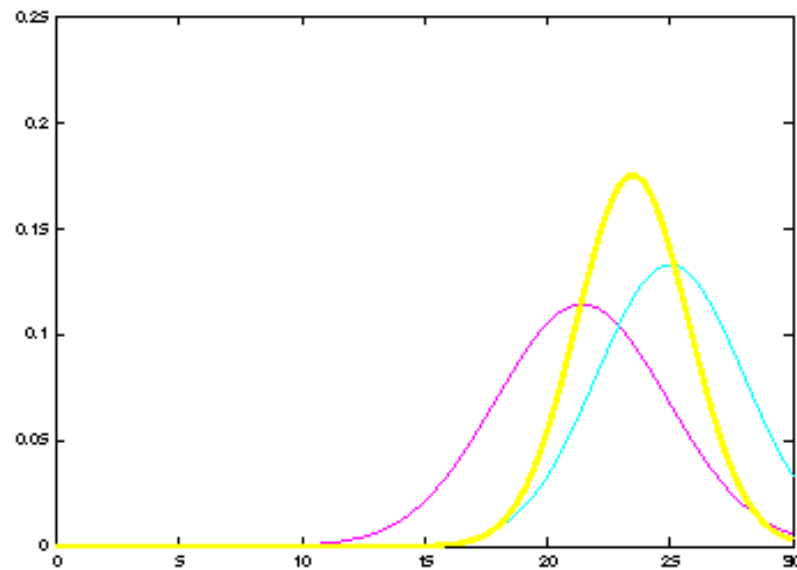
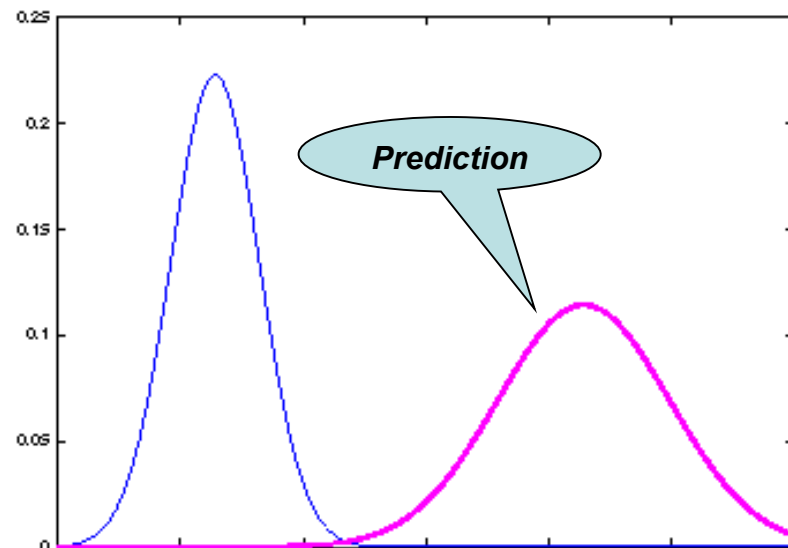
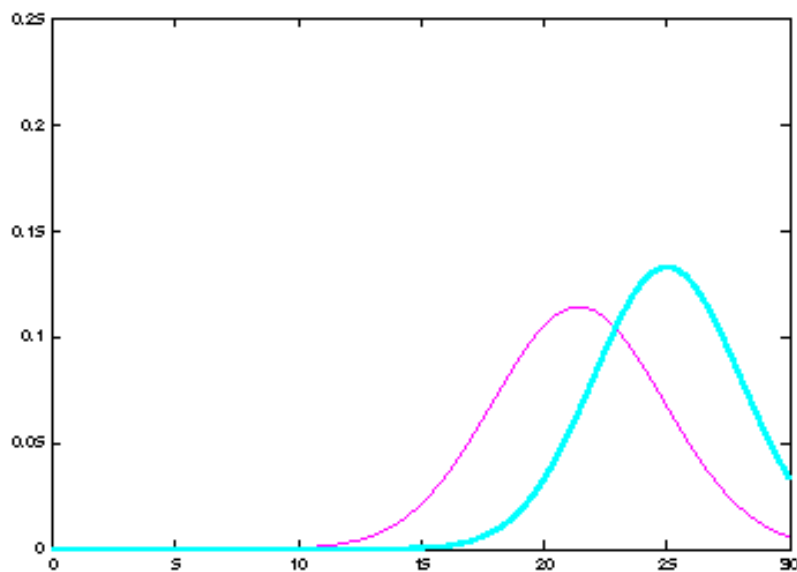
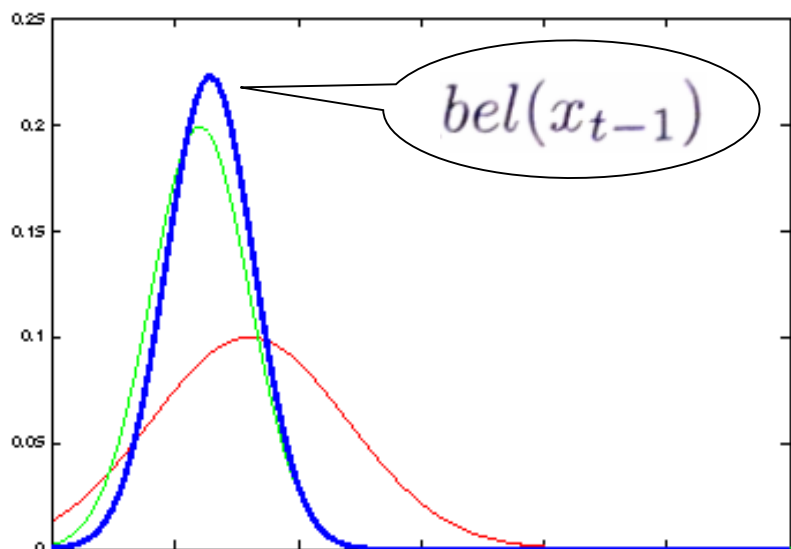


Gaussian Bayes Filter: Kalman

How does the Bayes Filter steps appear for a Linear-Gaussian System?



Kalman Filter “Prediction”: 1D



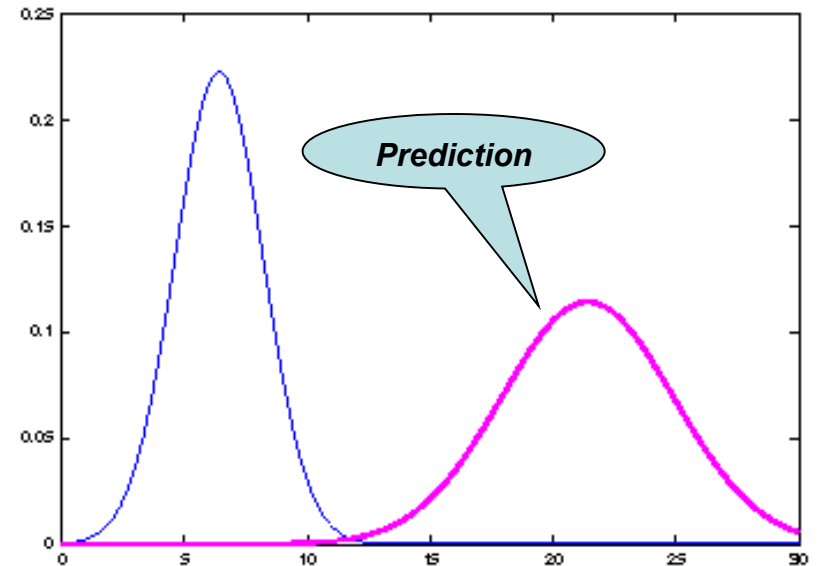


Kalman Filter “Prediction”

$$\overline{bel}(x_t) = \begin{cases} \bar{\mu}_t = a_t \mu_{t-1} + b_t u_t \\ \bar{\sigma}_t^2 = a_t^2 \sigma_t^2 + \sigma_{act,t}^2 \end{cases}$$

$$\overline{bel}(x_t) = \begin{cases} \bar{\mu}_t = A_t \mu_{t-1} + B_t u_t \\ \bar{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + R_t \end{cases}$$

Multivariate



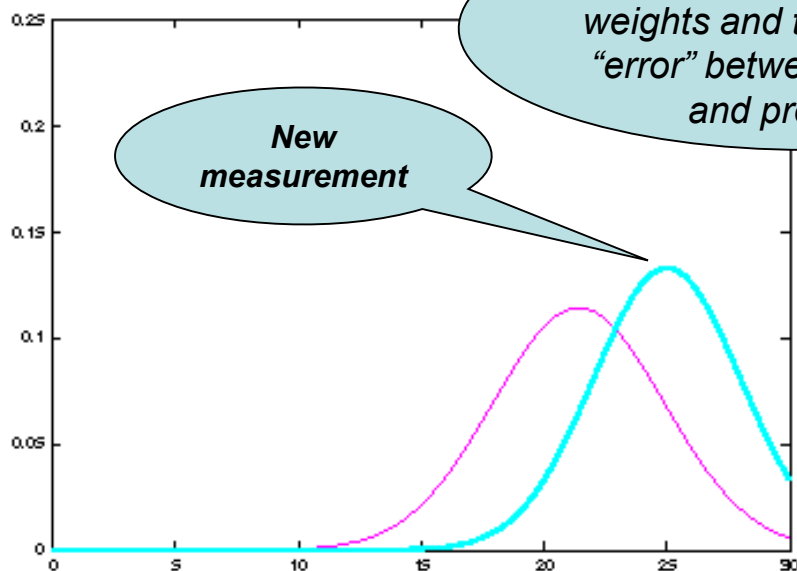


Kalman Filter Measurement Update

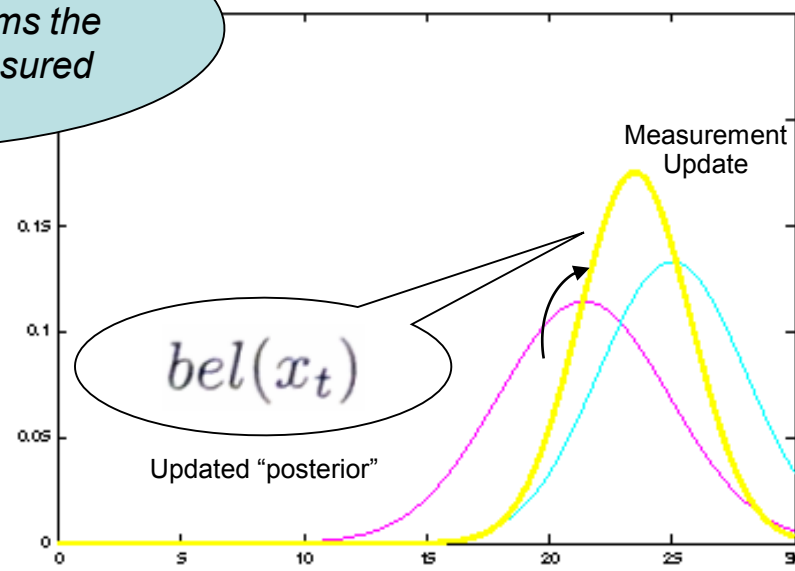
$$bel(x_t) = \begin{cases} \mu_t = \bar{\mu}_t + K_t(z_t - \bar{\mu}_t) \\ \sigma_t^2 = (1 - K_t)\bar{\sigma}_t^2 \end{cases} \quad \text{with} \quad K_t = \frac{\bar{\sigma}_t^2}{\bar{\sigma}_t^2 + \bar{\sigma}_{obs,t}^2}$$

$$bel(x_t) = \begin{cases} \mu_t = \bar{\mu}_t + K_t(z_t - C_t\bar{\mu}_t) \\ \Sigma_t = (I - K_tC_t)\bar{\Sigma}_t \end{cases} \quad \text{with} \quad K_t = \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1}$$

Multivariate



The "Kalman Gain" both weights and transforms the "error" between measured and predicted.





Kalman Filter Initialization

- Initial belief is also Normally (Gaussian) distributed:

$$bel(x_0) = N(x_0; \mu_0, \Sigma_0)$$



Linear Gaussian Systems: Dynamics

- Dynamics: Next state is a linear function of current state and control plus additive noise:

$$x_t = A_t x_{t-1} + B_t u_t + \varepsilon_t$$

$$p(x_t | u_t, x_{t-1}) = N(x_t; A_t x_{t-1} + B_t u_t, R_t)$$

$$\begin{array}{ccc} \overline{bel}(x_t) = \int p(x_t | u_t, x_{t-1}) & & bel(x_{t-1}) dx_{t-1} \\ \Downarrow & & \Downarrow \\ \sim N(x_t; A_t x_{t-1} + B_t u_t, R_t) & \sim & N(x_{t-1}; \mu_{t-1}, \Sigma_{t-1}) \end{array}$$



Linear Gaussian Systems: Dynamics

$$\overline{bel}(x_t) = \int p(x_t | u_t, x_{t-1}) \quad bel(x_{t-1}) dx_{t-1}$$



$$\sim N(x_t; A_t x_{t-1} + B_t u_t, R_t) \quad \sim N(x_{t-1}; \mu_{t-1}, \Sigma_{t-1})$$



$$\overline{bel}(x_t) = \eta \int \exp \left\{ -\frac{1}{2} (x_t - A_t x_{t-1} - B_t u_t)^T R_t^{-1} (x_t - A_t x_{t-1} - B_t u_t) \right\} \\ \exp \left\{ -\frac{1}{2} (x_{t-1} - \mu_{t-1})^T \Sigma_{t-1}^{-1} (x_{t-1} - \mu_{t-1}) \right\} dx_{t-1}$$

$$\overline{bel}(x_t) = \begin{cases} \bar{\mu}_t = A_t \mu_{t-1} + B_t u_t \\ \bar{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + R_t \end{cases}$$



**After a non-trivial
mathematical derivation**



Linear Gaussian Systems: Observations

- Observations are linear function of state plus additive noise:

$$z_t = C_t x_t + \delta_t$$

$$p(z_t | x_t) = N(z_t; C_t x_t, Q_t)$$

$$\begin{array}{ccc} \text{bel}(x_t) = & \eta & p(z_t | x_t) & \overline{\text{bel}}(x_t) \\ & & \Downarrow & \Downarrow \\ & & \sim N(z_t; C_t x_t, Q_t) & \sim N(x_t; \bar{\mu}_t, \bar{\Sigma}_t) \end{array}$$



Linear Gaussian Systems: Observations

$$\begin{aligned} bel(x_t) &= \eta \quad p(z_t | x_t) & \overline{bel}(x_t) \\ &\Downarrow & \Downarrow \\ &\sim N(z_t; C_t x_t, Q_t) & \sim N(x_t; \bar{\mu}_t, \bar{\Sigma}_t) \\ &\Downarrow \\ bel(x_t) &= \eta \exp \left\{ -\frac{1}{2} (z_t - C_t x_t)^T Q_t^{-1} (z_t - C_t x_t) \right\} \exp \left\{ -\frac{1}{2} (x_t - \bar{\mu}_t)^T \bar{\Sigma}_t^{-1} (x_t - \bar{\mu}_t) \right\} \\ \\ bel(x_t) &= \begin{cases} \mu_t = \bar{\mu}_t + K_t (z_t - C_t \bar{\mu}_t) \\ \Sigma_t = (I - K_t C_t) \bar{\Sigma}_t \end{cases} & \text{with } K_t = \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1} \end{aligned}$$

***After a non-trivial
mathematical derivation***



The Kalman Filter Algorithm

1: **Algorithm** Kalman_filter($\mu_{t-1}, \Sigma_{t-1}, u_t, z_t$):

2: *Prediction:*

3: $\bar{\mu}_t = A_t \mu_{t-1} + B_t u_t$

4: $\bar{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + R_t$

5: *Measurement Update:*

6: $K_t = \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1}$

7: $\mu_t = \bar{\mu}_t + K_t (z_t - C_t \bar{\mu}_t)$

8: $\Sigma_t = (I - K_t C_t) \bar{\Sigma}_t$

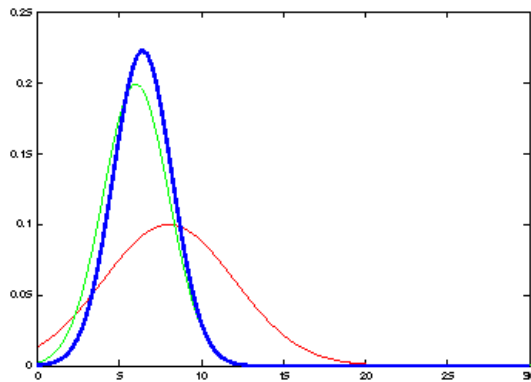
9: return μ_t, Σ_t

“Kalman Gain”

“Innovation”
(The difference between the
predicted and actual
measurement)

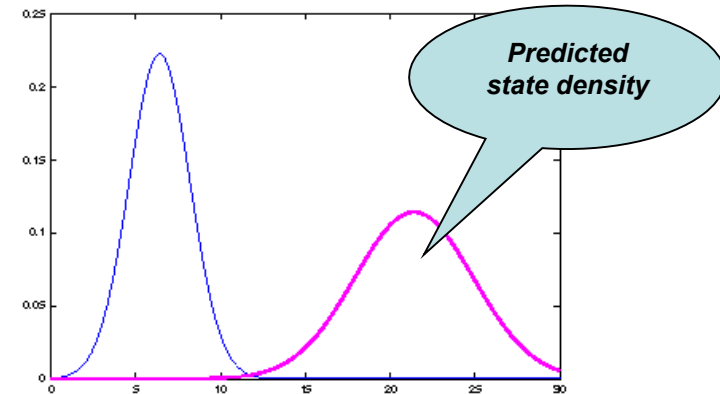


The Prediction-Correction Cycle



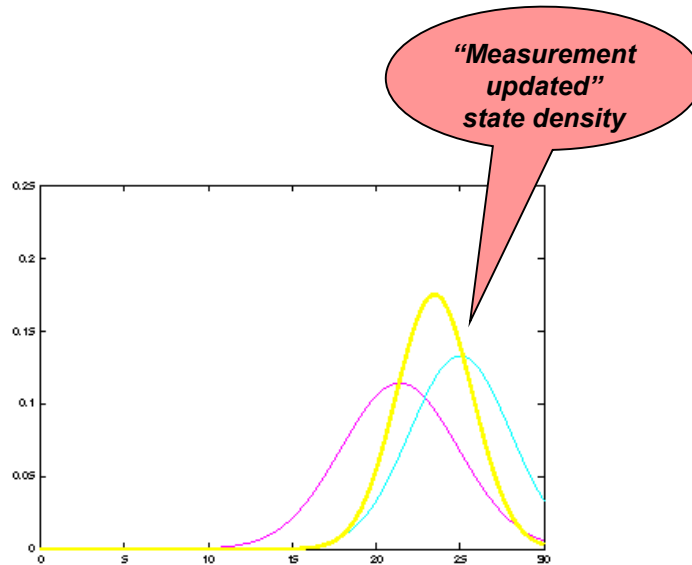
$$\overline{bel}(x_t) = \begin{cases} \bar{\mu}_t = a_t \mu_{t-1} + b_t u_t \\ \bar{\sigma}_t^2 = a_t^2 \sigma_t^2 + \sigma_{act,t}^2 \end{cases}$$

$$\overline{bel}(x_t) = \begin{cases} \bar{\mu}_t = A_t \mu_{t-1} + B_t u_t \\ \bar{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + R_t \end{cases}$$



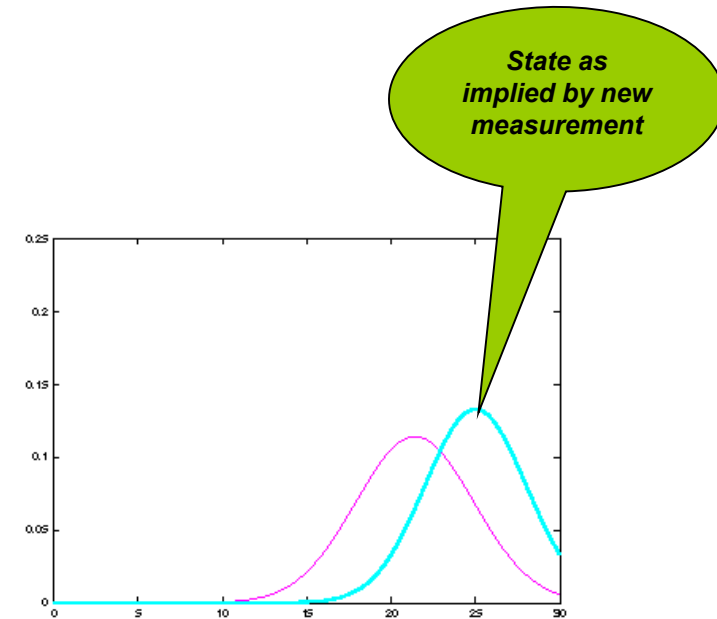


The Prediction-Correction Cycle



$$bel(x_t) = \begin{cases} \mu_t = \bar{\mu}_t + K_t(z_t - \bar{\mu}_t) \\ \sigma_t^2 = (1 - K_t)\bar{\sigma}_t^2, K_t = \frac{\bar{\sigma}_t^2}{\bar{\sigma}_t^2 + \bar{\sigma}_{obs,t}^2} \end{cases}$$

$$bel(x_t) = \begin{cases} \mu_t = \bar{\mu}_t + K_t(z_t - C_t\bar{\mu}_t) \\ \Sigma_t = (I - K_tC_t)\bar{\Sigma}_t, K_t = \bar{\Sigma}_tC_t^T(C_t\bar{\Sigma}_tC_t^T + Q_t)^{-1} \end{cases}$$



Correction
(Measurement Update)



The Prediction-Correction Cycle



Prediction

$$bel(x_t) = \begin{cases} \mu_t = \bar{\mu}_t + K_t(z_t - \bar{\mu}_t) \\ \sigma_t^2 = (1 - K_t)\bar{\sigma}_t^2 \end{cases}, K_t = \frac{\bar{\sigma}_t^2}{\bar{\sigma}_t^2 + \bar{\sigma}_{obs,t}^2}$$

$$bel(x_t) = \begin{cases} \mu_t = \bar{\mu}_t + K_t(z_t - C_t\bar{\mu}_t) \\ \Sigma_t = (I - K_tC_t)\bar{\Sigma}_t \end{cases}, K_t = \bar{\Sigma}_tC_t^T(C_t\bar{\Sigma}_tC_t^T + Q_t)^{-1}$$

$$\overline{bel}(x_t) = \begin{cases} \bar{\mu}_t = a_t\mu_{t-1} + b_tu_t \\ \bar{\sigma}_t^2 = a_t^2\sigma_{t-1}^2 + \sigma_{act,t}^2 \end{cases}$$

$$\overline{bel}(x_t) = \begin{cases} \bar{\mu}_t = A_t\mu_{t-1} + B_tu_t \\ \bar{\Sigma}_t = A_t\Sigma_{t-1}A_t^T + R_t \end{cases}$$



Correction
(Measurement Update)



The Kalman Filter Summary

- *Highly efficient*: Polynomial in measurement dimensionality k and state dimensionality n :

$$O(k^{2.376} + n^2)$$

- *Optimal for linear Gaussian systems!*
- *Uni-modal representation.*

But...

- Most robotics systems are **nonlinear**!
- How can we handle nonlinearity?



Nonlinear Dynamic Systems

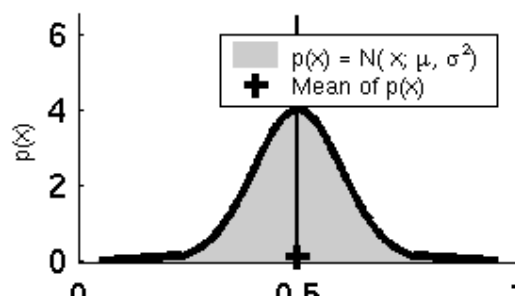
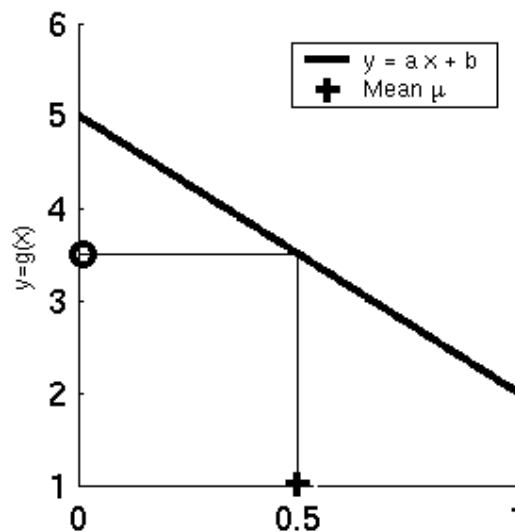
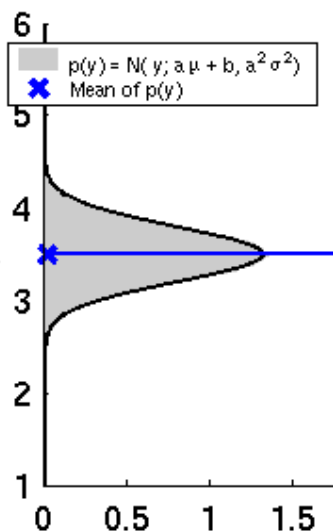
- Most realistic robotic problems involve nonlinear functions

$$x_t = g(u_t, x_{t-1})$$

$$z_t = h(x_t)$$



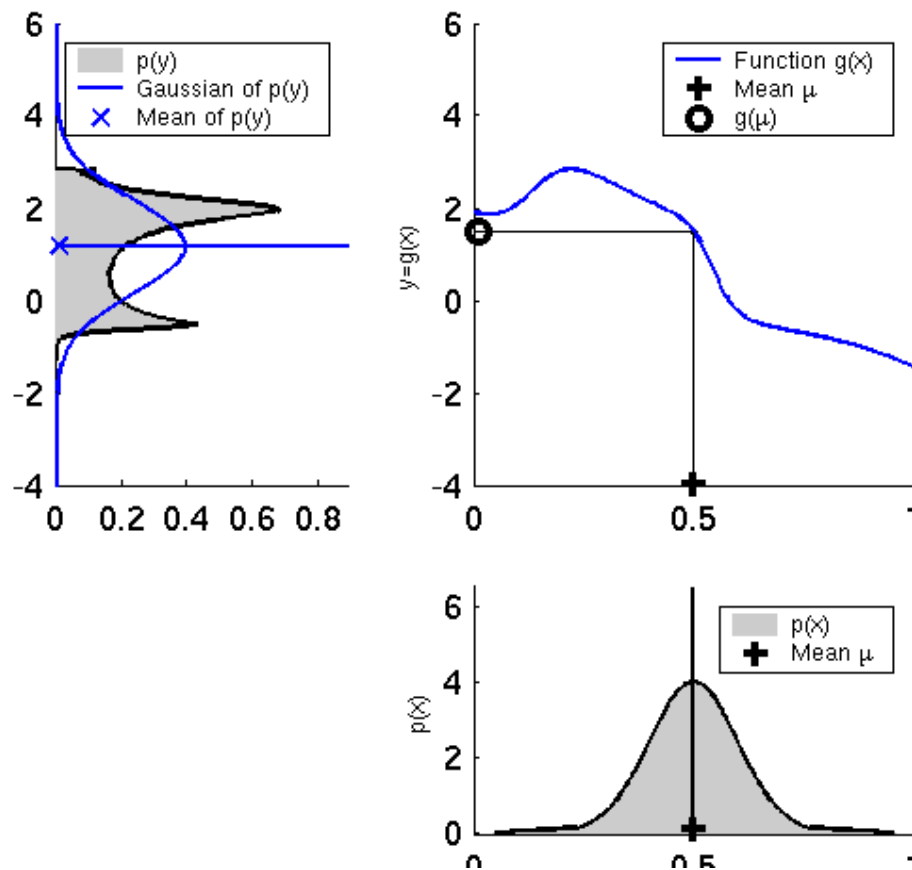
Linearity Assumption Revisited



- Linearity $\rightarrow$ Gaussian in, Gaussian Out...



Nonlinear Function



- Non-Linearity $\rightarrow$ Output no longer Gaussian!

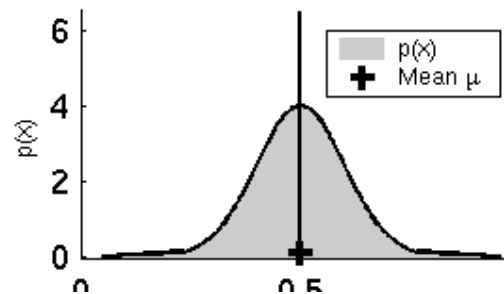
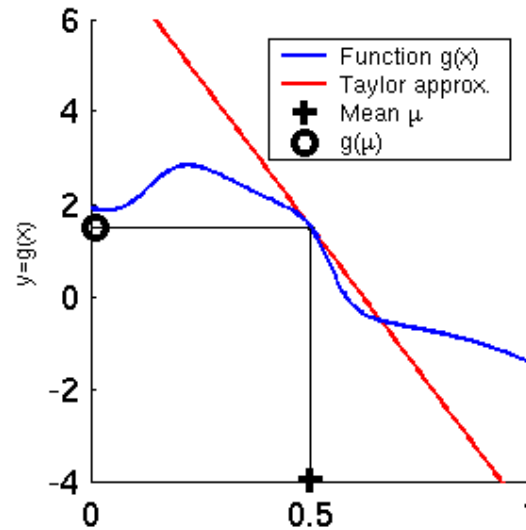
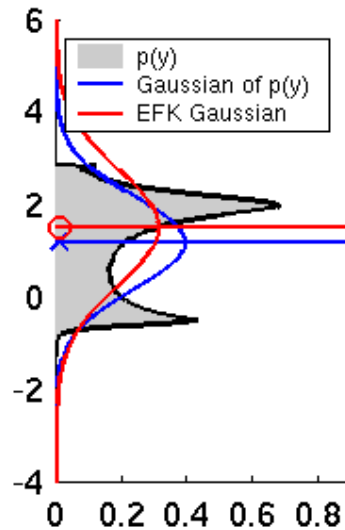


Nonlinear Function

- **Question:**
Can we find a linear approximation to the function?
- **Yes. We can find an approximation around the current state estimate point.**
 - Approximation through *Taylor Series Expansion* → **Extended Kalman Filter (EKF)**
 - Approximation through Unscented Transform → **Unscented Kalman Filter (UKF)**



EKF Linearization (Taylor Series) 1

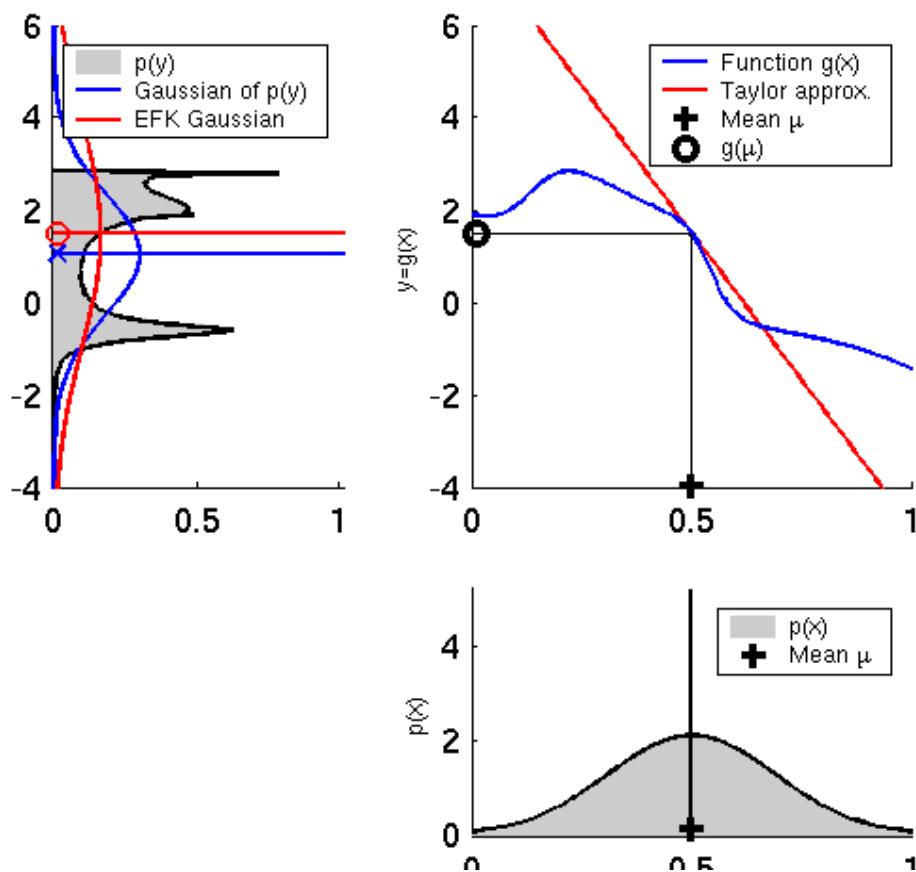


$$f(x) \cong f(x^*) + \left. \frac{\partial f(x)}{\partial x} \right|_{x=x^*} (x - x^*)$$

How accurate is this approximation?



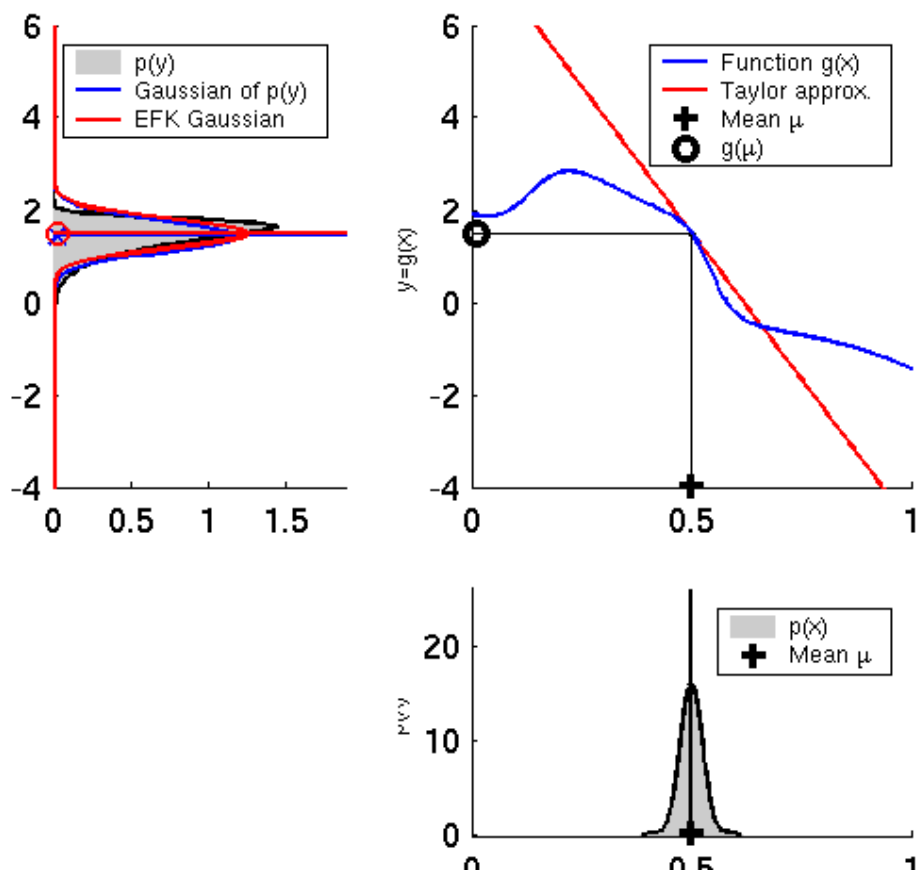
EKF Linearization (Taylor Series) 2



Amount of uncertainty affects the linearization performance!!



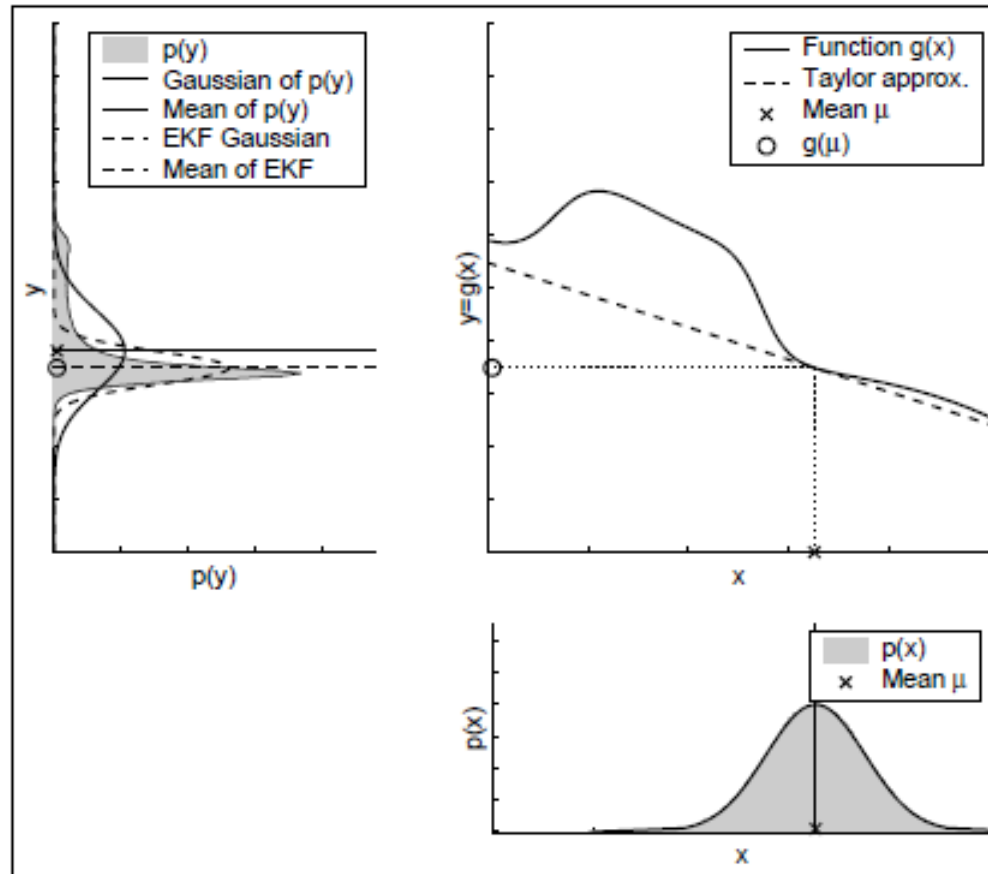
EKF Linearization (Taylor Series) 3



Amount of uncertainty affects the linearization performance!!



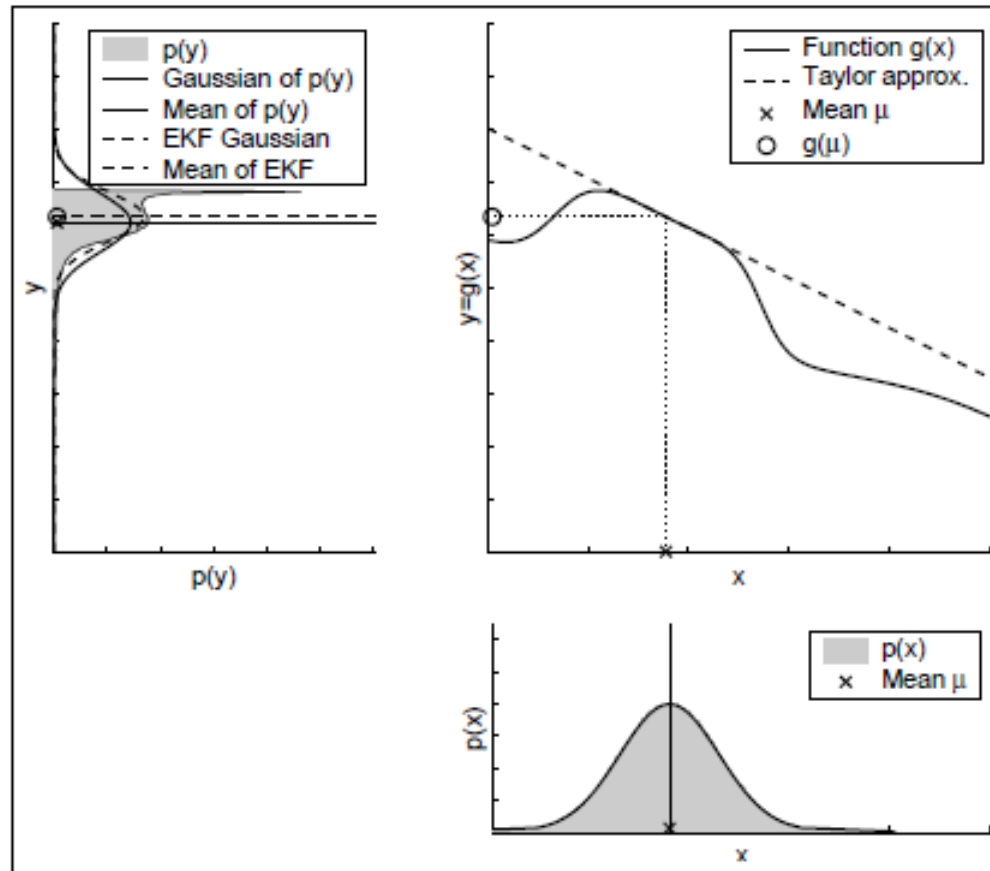
EKF Linearization (Taylor Series) 4



Amount of nonlinearity affects linearization performance!!



EKF Linearization (Taylor Series) 5



Amount of nonlinearity affects linearization performance!!



EKF Linearization

First Order Taylor Series Expansion

- State Transition:*

$$g(u_t, x_{t-1}) \approx g(u_t, \mu_{t-1}) + \left. \frac{\partial g(u_t, x_{t-1})}{\partial x_{t-1}} \right|_{x_{t-1}=\mu_{t-1}} (x_{t-1} - \mu_{t-1})$$

$$g(u_t, x_{t-1}) \approx g(u_t, \mu_{t-1}) + G_t (x_{t-1} - \mu_{t-1})$$

$$G_t = \frac{\partial g(u_t, x_{t-1})}{\partial x_{t-1}} = \begin{pmatrix} \frac{\partial x'}{\partial x_{t-1,x}} & \frac{\partial x'}{\partial x_{t-1,y}} & \frac{\partial x'}{\partial x_{t-1,\theta}} \\ \frac{\partial y'}{\partial x_{t-1,x}} & \frac{\partial y'}{\partial x_{t-1,y}} & \frac{\partial y'}{\partial x_{t-1,\theta}} \\ \frac{\partial \theta'}{\partial x_{t-1,x}} & \frac{\partial \theta'}{\partial x_{t-1,y}} & \frac{\partial \theta'}{\partial x_{t-1,\theta}} \end{pmatrix}$$

- Measurement:*

$$h(x_t) \approx h(\bar{\mu}_t) + \left. \frac{\partial h(x_t)}{\partial x_t} \right|_{x_t=\bar{\mu}_t} (x_t - \bar{\mu}_t)$$

$$h(x_t) \approx h(\bar{\mu}_t) + H_t (x_t - \bar{\mu}_t)$$

The **Jacobian** matrix of the function $g(.)$



The EKF Algorithm

1. **Extended_Kalman_filter**($\mu_{t-1}, \Sigma_{t-1}, u_t, z_t$):

2. **Prediction:**

3. $\bar{\mu}_t = g(u_t, \mu_{t-1})$

4. $\bar{\Sigma}_t = G_t \Sigma_{t-1} G_t^T + R_t$

5. **Correction:**

6. $K_t = \bar{\Sigma}_t H_t^T (H_t \bar{\Sigma}_t H_t^T + Q_t)^{-1}$

7. $\mu_t = \bar{\mu}_t + K_t (z_t - h(\bar{\mu}_t))$

8. $\Sigma_t = (I - K_t H_t) \bar{\Sigma}_t$

9. **Return** μ_t, Σ_t

$$\bar{\mu}_t = A_t \mu_{t-1} + B_t u_t$$

$$\bar{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + R_t$$

$$K_t = \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1}$$

$$\mu_t = \bar{\mu}_t + K_t (z_t - C_t \bar{\mu}_t)$$

$$\Sigma_t = (I - K_t C_t) \bar{\Sigma}_t$$

$$G_t = \left. \frac{\partial g(u_t, x_{t-1})}{\partial x_{t-1}} \right|_{x_{t-1} = \mu_{t-1}}$$

$$H_t = \left. \frac{\partial h(x_t)}{\partial x_t} \right|_{x_t = \bar{\mu}_t}$$

Compare with the
Kalman Filter



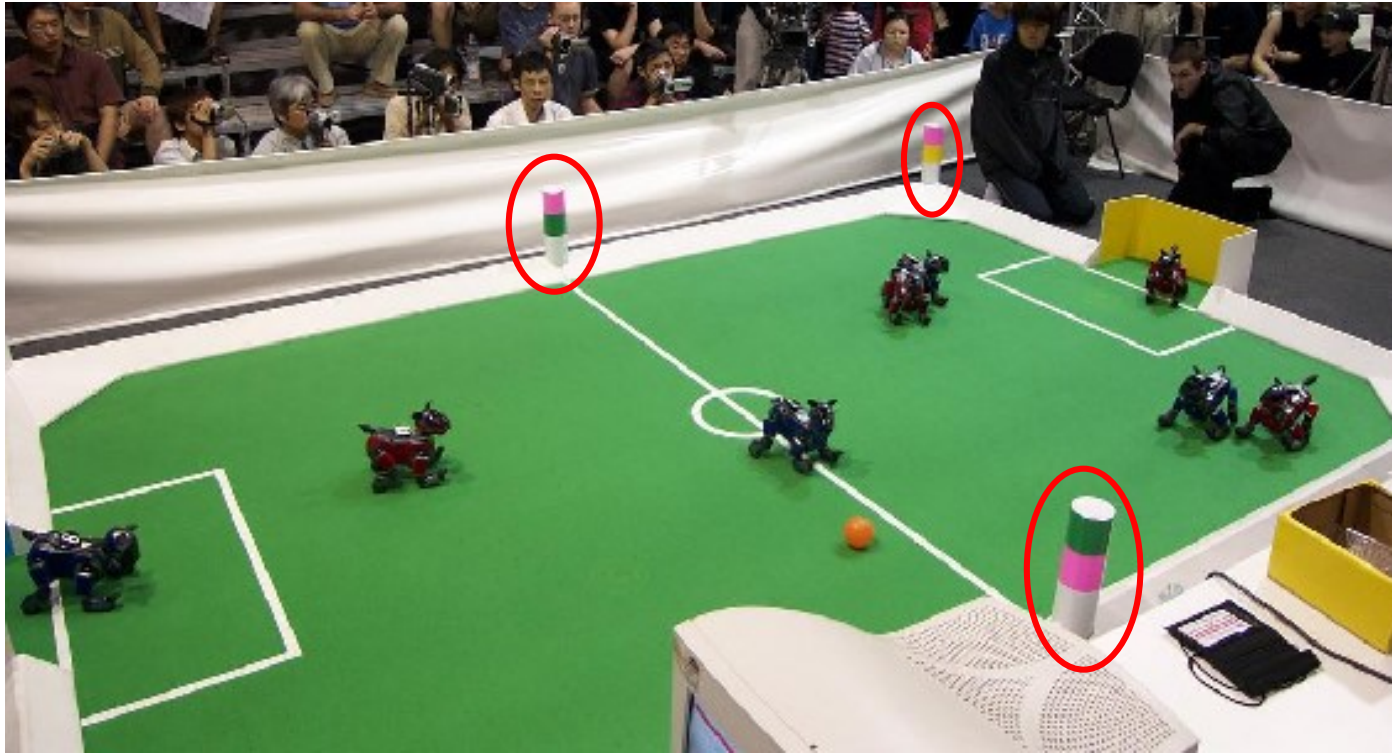
EKF Localization

“Using sensory information to locate the robot in its environment is the most fundamental problem to providing a mobile robot with autonomous capabilities.” [Cox '91]

- **Given**
 - Map of the environment.
 - Sequence of sensor measurements.
- **Wanted**
 - Estimate of the robot's position.
- **Problem classes**
 - Position tracking
 - Global localization
 - Kidnapped robot problem (recovery)



Landmark Based Localization





EKF Localization

(Known Correspondence)

1: Algorithm EKF_localization_known_correspondences($\mu_{t-1}, \Sigma_{t-1}, u_t, z_t, c_t, m$):

2: $\theta = \mu_{t-1, \theta}$

$$3: G_t = \begin{pmatrix} 1 & 0 & -\frac{v_t}{\omega_t} \cos \theta + \frac{v_t}{\omega_t} \cos(\theta + \omega_t \Delta t) \\ 0 & 1 & -\frac{v_t}{\omega_t} \sin \theta + \frac{v_t}{\omega_t} \sin(\theta + \omega_t \Delta t) \\ 0 & 0 & 1 \end{pmatrix}$$

$\frac{\partial g(u_t, x_{t-1})}{\partial x_{t-1}} \Big|_{x_{t-1}=\mu_{t-1}}$
 $\frac{\partial g(u_t, x_{t-1})}{\partial u_t} \Big|_{x_{t-1}=\mu_{t-1}, u_t=u_t}$

$$4: V_t = \begin{pmatrix} -\frac{\sin \theta + \sin(\theta + \omega_t \Delta t)}{\omega_t} & \frac{v_t(\sin \theta - \sin(\theta + \omega_t \Delta t))}{\omega_t^2} + \frac{v_t \cos(\theta + \omega_t \Delta t) \Delta t}{\omega_t} \\ \frac{\cos \theta - \cos(\theta + \omega_t \Delta t)}{\omega_t} & -\frac{v_t(\cos \theta - \cos(\theta + \omega_t \Delta t))}{\omega_t^2} + \frac{v_t \sin(\theta + \omega_t \Delta t) \Delta t}{\omega_t} \\ 0 & \Delta t \end{pmatrix}$$

$$5: M_t = \begin{pmatrix} \alpha_1 v_t^2 + \alpha_2 \omega_t^2 & 0 \\ 0 & \alpha_3 v_t^2 + \alpha_4 \omega_t^2 \end{pmatrix}$$

$$6: \bar{\mu}_t = \mu_{t-1} + \begin{pmatrix} -\frac{v_t}{\omega_t} \sin \theta + \frac{v_t}{\omega_t} \sin(\theta + \omega_t \Delta t) \\ \frac{v_t}{\omega_t} \cos \theta - \frac{v_t}{\omega_t} \cos(\theta + \omega_t \Delta t) \\ \omega_t \Delta t \end{pmatrix}$$

$$7: \bar{\Sigma}_t = G_t \Sigma_{t-1} G_t^T + V_t M_t V_t^T$$

Motion "prediction" step



EKF Localization

(Remember Landmark Based Measurement Model)

For the Measurement “update” step:

Consider a Landmark Based Measurement Model with known correspondence

```
1:  Algorithm landmark_model_known_correspondence( $f_t^i, c_t^i, x_t, m$ ):  
2:       $j = c_t^i$   
3:       $\hat{r} = \sqrt{(m_{j,x} - x)^2 + (m_{j,y} - y)^2}$   
4:       $\hat{\phi} = \text{atan2}(m_{j,y} - y, m_{j,x} - x)$   
5:       $q = \text{prob}(r_t^i - \hat{r}, \sigma_r) \cdot \text{prob}(\phi_t^i - \hat{\phi}, \sigma_\phi) \cdot \text{prob}(s_t^i - s_j, \sigma_s)$   
6:      return  $q$ 
```

observed feature $f_t^i = (r_t^i \ \phi_t^i \ s_t^i)^T$

the robot pose $x_t = (x \ y \ \theta)^T$



EKF Localization (Known Correspondence)

8: $Q_t = \begin{pmatrix} \sigma_r^2 & 0 & 0 \\ 0 & \sigma_\phi^2 & 0 \\ 0 & 0 & \sigma_s^2 \end{pmatrix}$
 9: for all observed features $z_t^i = (r_t^i \ \phi_t^i \ s_t^i)^T$ do

10: $j = c_t^i$

11: $q = (m_{j,x} - \bar{\mu}_{t,x})^2 + (m_{j,y} - \bar{\mu}_{t,y})^2$

12: $\hat{z}_t^i = \begin{pmatrix} \sqrt{q} \\ \text{atan2}(m_{j,y} - \bar{\mu}_{t,y}, m_{j,x} - \bar{\mu}_{t,x}) - \bar{\mu}_{t,\theta} \\ m_{j,s} \end{pmatrix}$

13: $H_t^i = \begin{pmatrix} -\frac{m_{j,x} - \bar{\mu}_{t,x}}{\sqrt{q}} & -\frac{m_{j,y} - \bar{\mu}_{t,y}}{\sqrt{q}} & 0 \\ \frac{m_{j,y} - \bar{\mu}_{t,y}}{q} & -\frac{m_{j,x} - \bar{\mu}_{t,x}}{q} & -1 \\ 0 & 0 & 0 \end{pmatrix}$

14: $S_t^i = H_t^i \bar{\Sigma}_t [H_t^i]^T + Q_t$

15: $K_t^i = \bar{\Sigma}_t [H_t^i]^T [S_t^i]^{-1}$

16: $\bar{\mu}_t = \bar{\mu}_t + K_t^i (z_t^i - \hat{z}_t^i)$

17: $\bar{\Sigma}_t = (I - K_t^i H_t^i) \bar{\Sigma}_t$

18: endfor

19: $\mu_t = \bar{\mu}_t$

20: $\Sigma_t = \bar{\Sigma}_t$

21: $p_{z_t} = \prod_i \det(2\pi S_t^i)^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (z_t^i - \hat{z}_t^i)^T [S_t^i]^{-1} (z_t^i - \hat{z}_t^i) \right\}$

22: return μ_t, Σ_t, p_{z_t}

All features are assumed independent and processed as if they were individual measurements (with no motion in between them)!

$$\left. \frac{\partial h(x_t, j, m)}{\partial x_t} \right|_{x_t = \bar{\mu}_t}$$

**Measurement
"update" step**

Independence of features used to compute a "measurement likelihood"



EKF Prediction Step

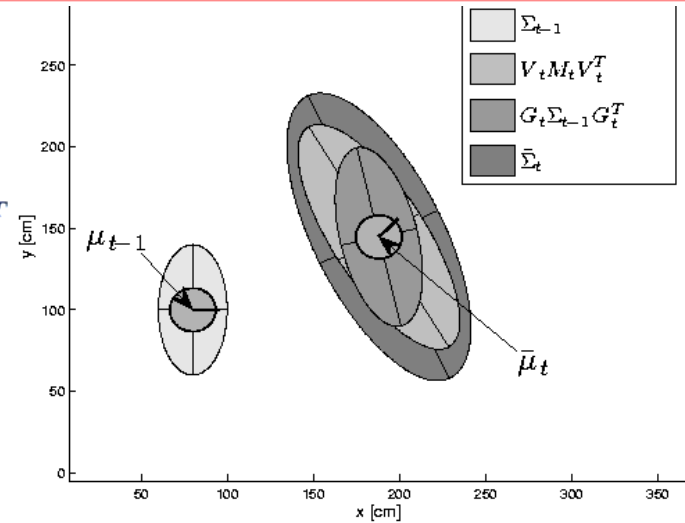
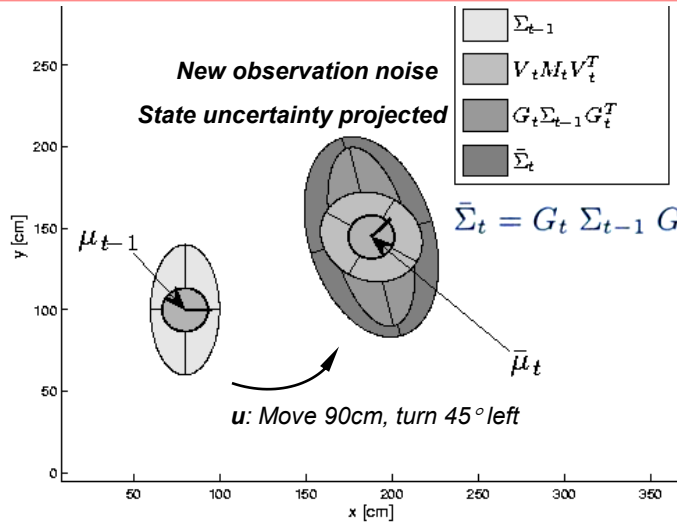
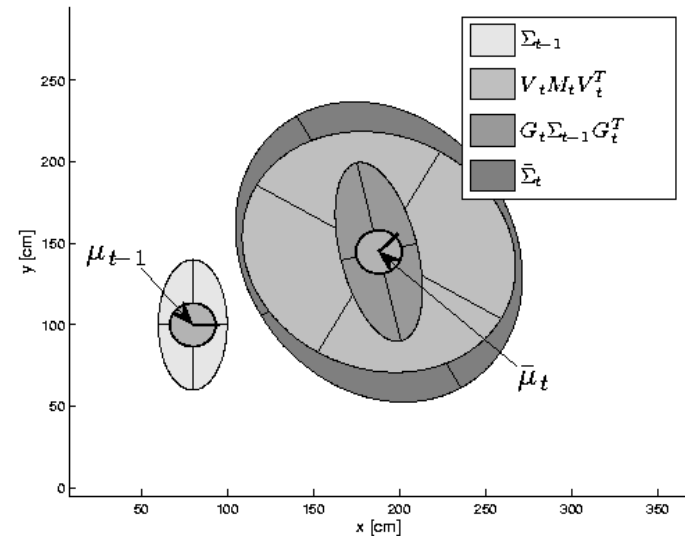
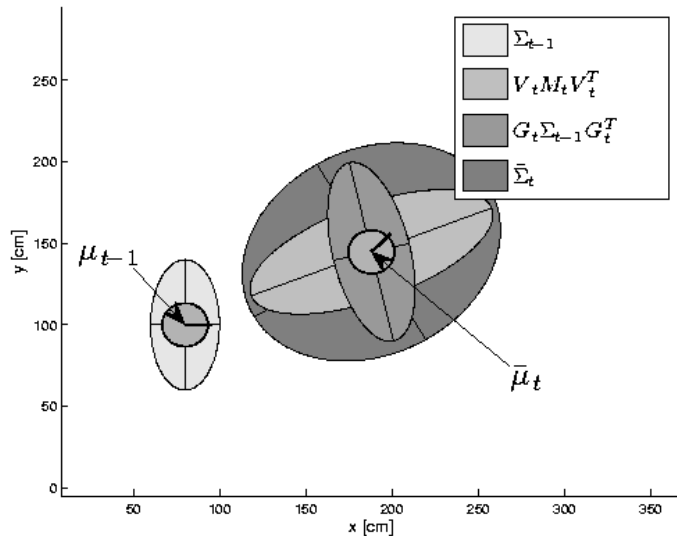
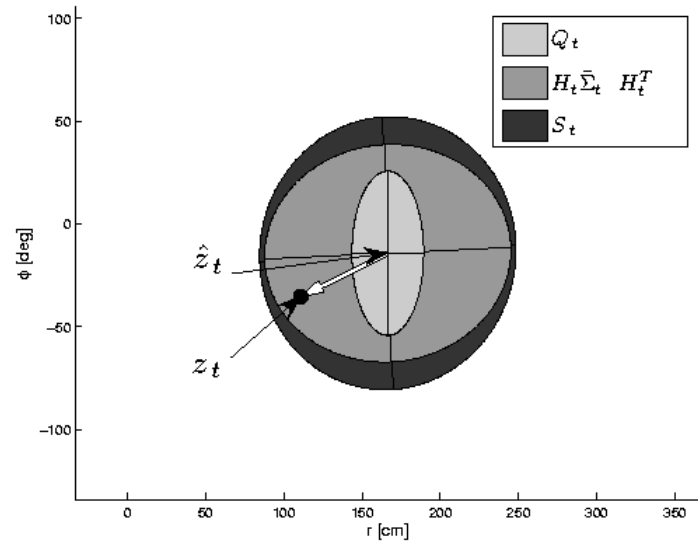
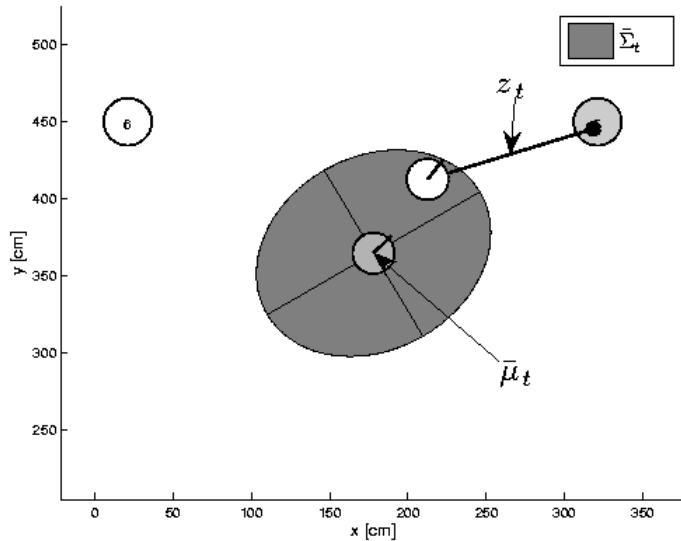
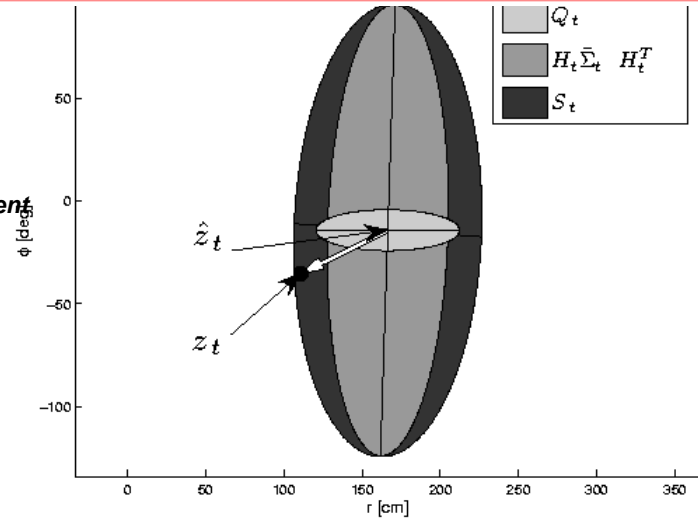
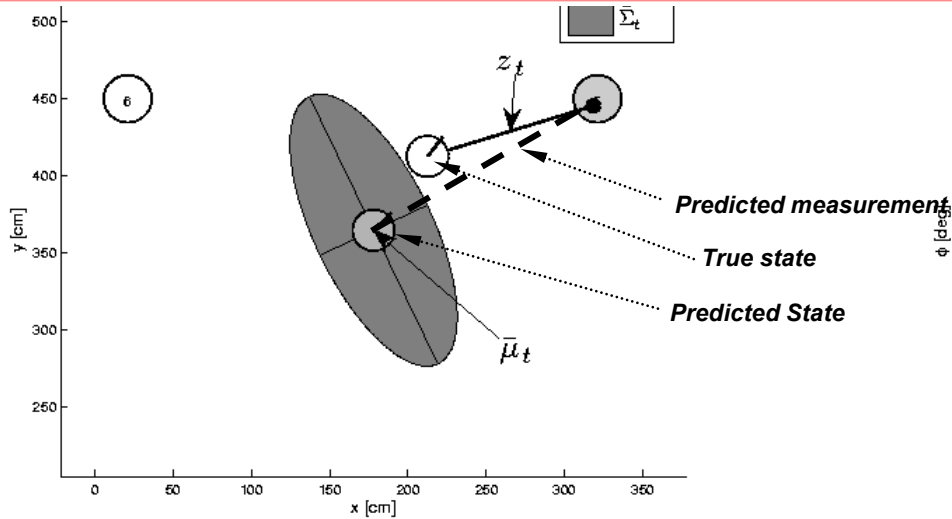


Illustration with different motion noise parameters



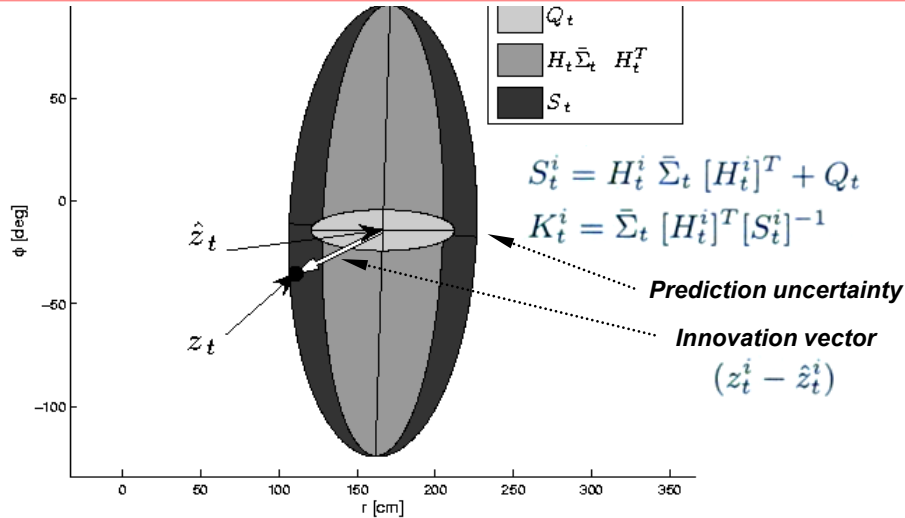


EKF Measurement Prediction

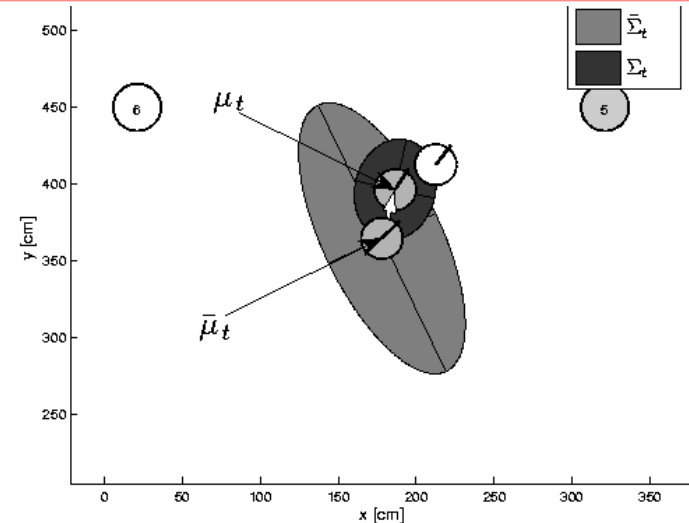




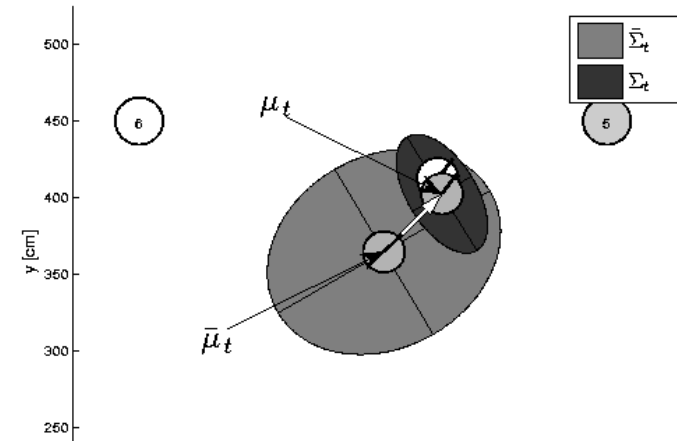
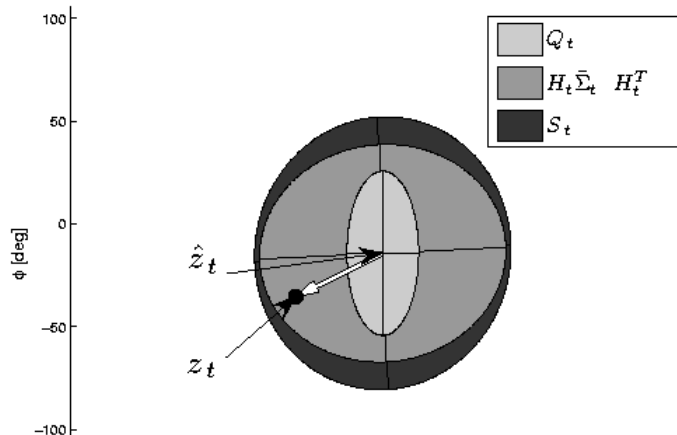
EKF Measurement Correction Step



Measurement predictions



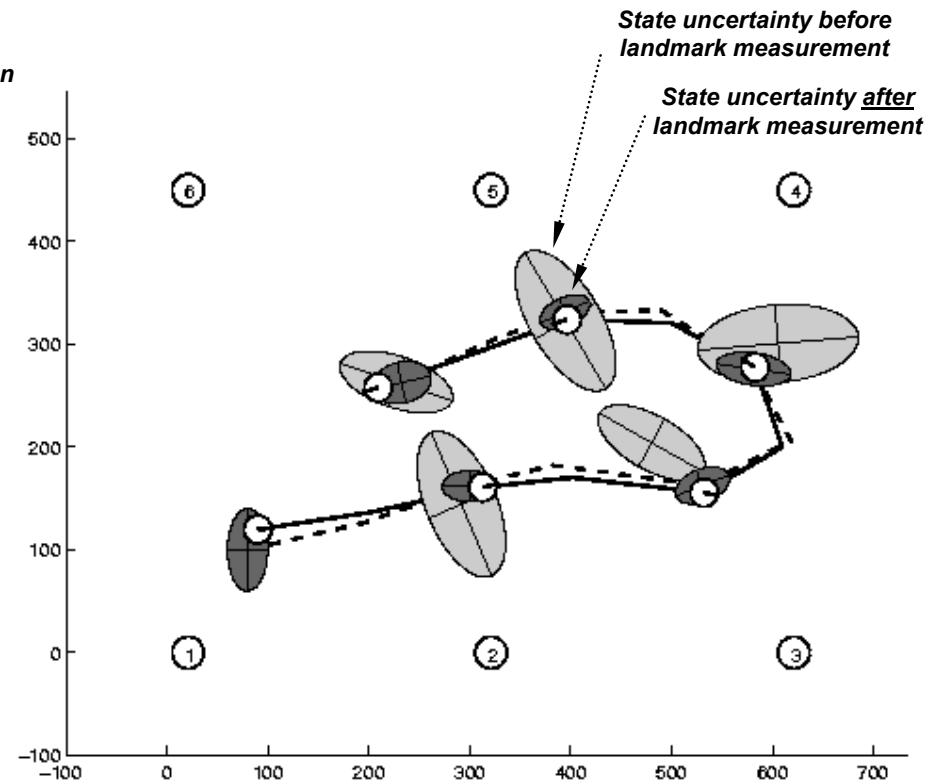
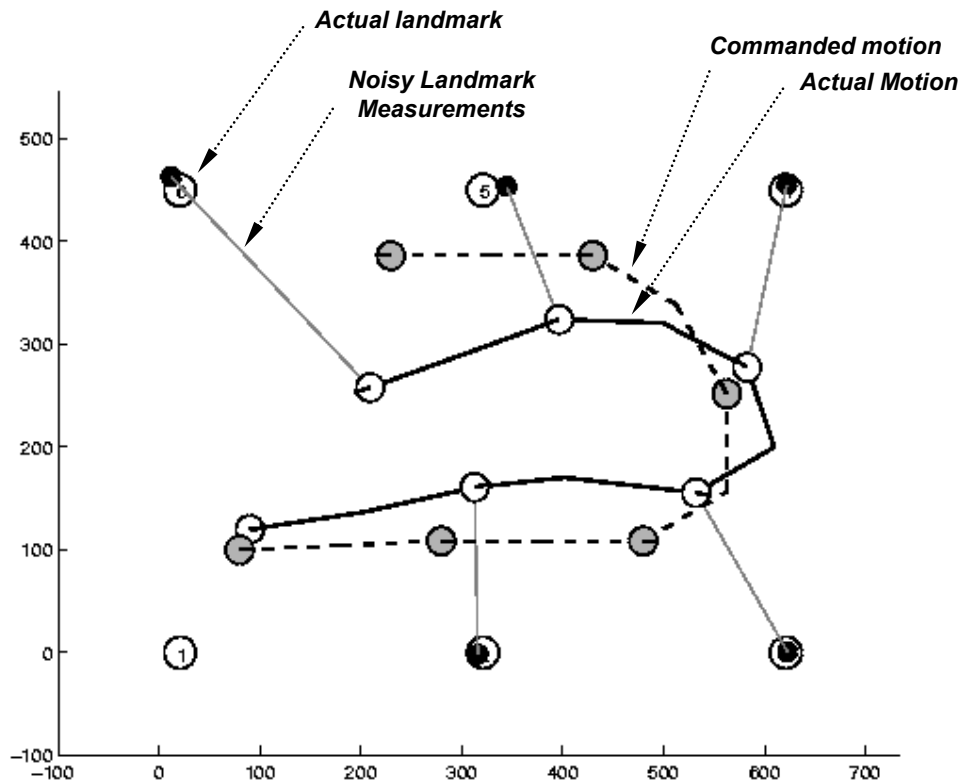
Resulting update to mean and covariance



The more certain the measurement prediction, the larger the Kalman Gain, hence the correction based on the projected innovation vector.



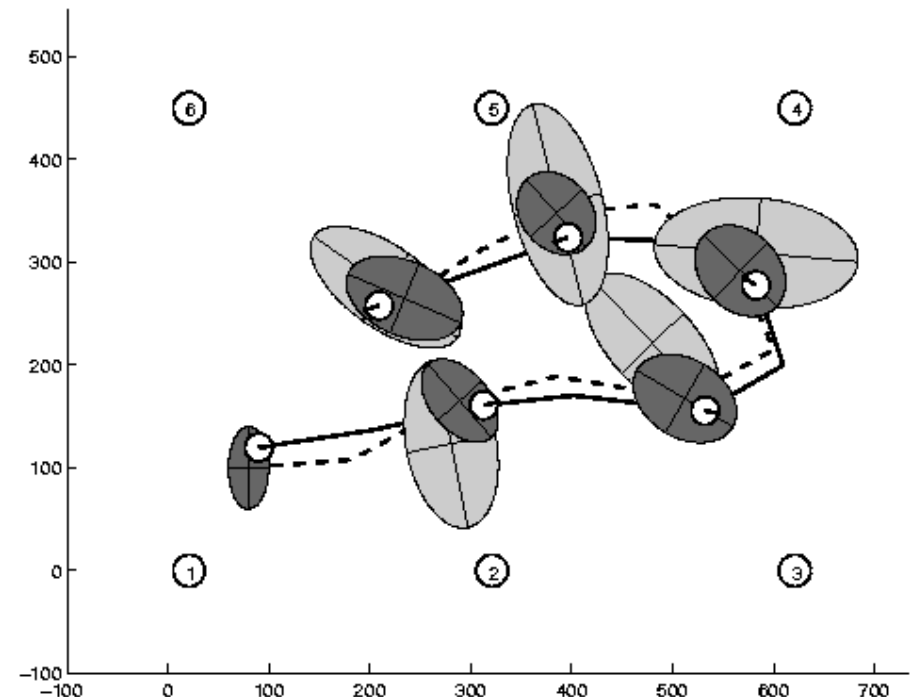
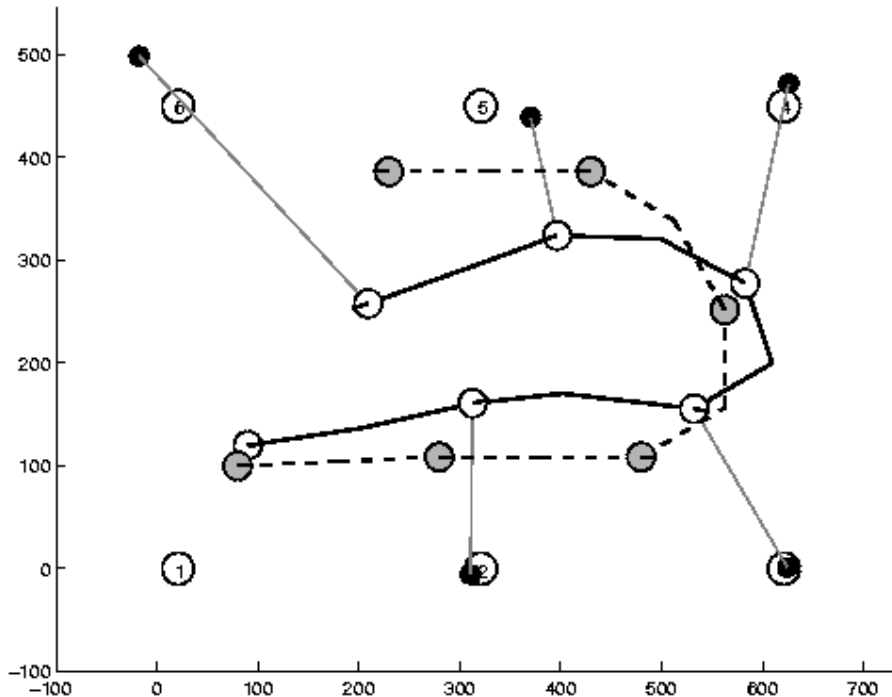
Estimation Sequence (Accurate Sensor)





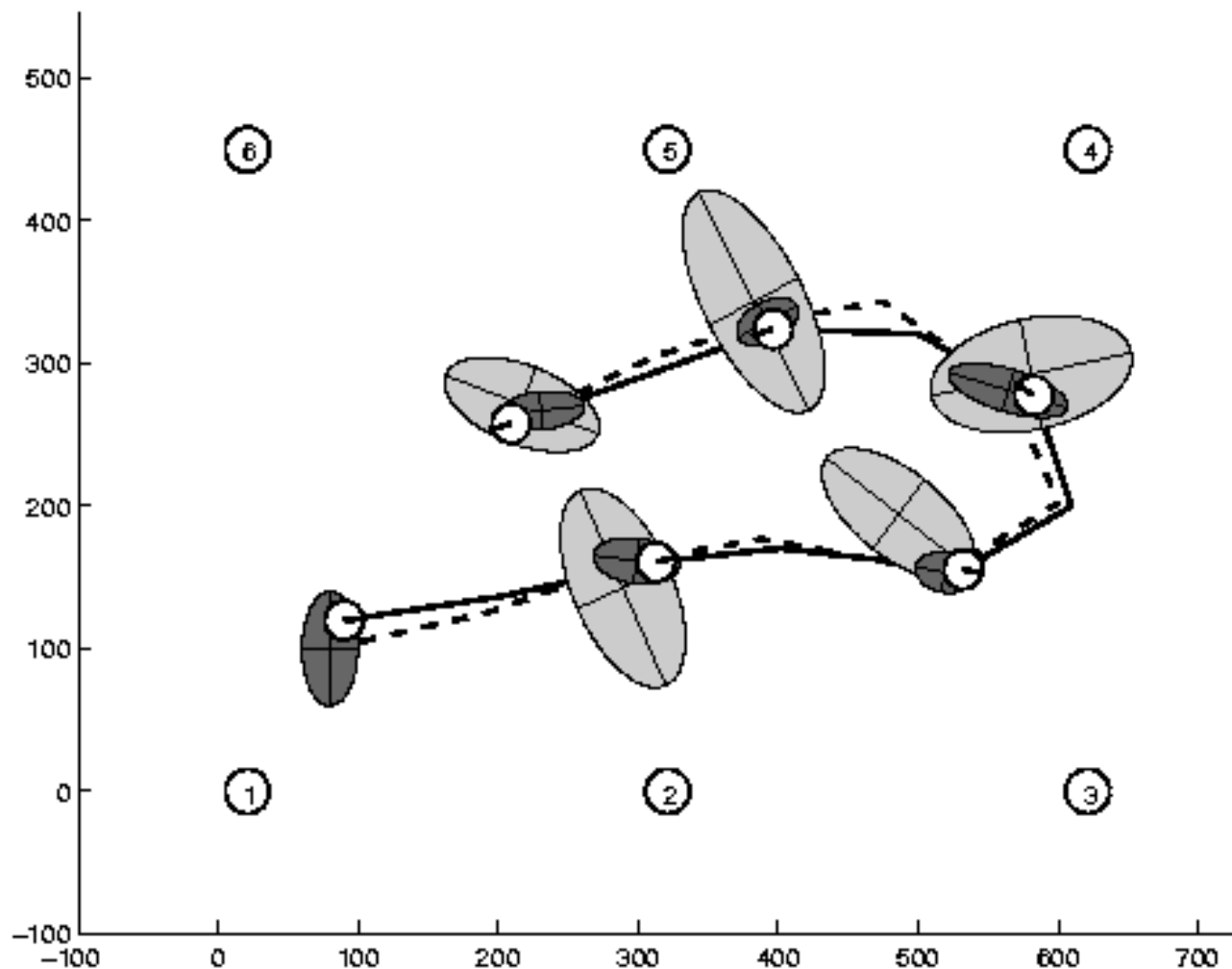
Estimation Sequence

(Less Accurate Sensor)





Comparison to Ground Truth



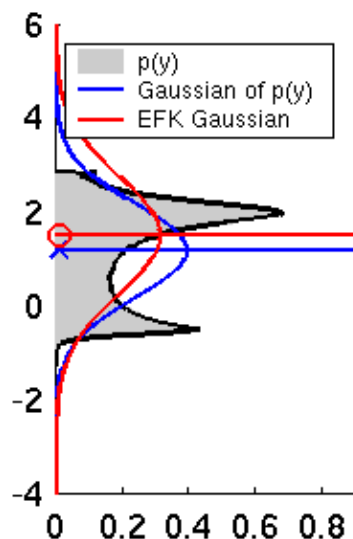


EKF Summary

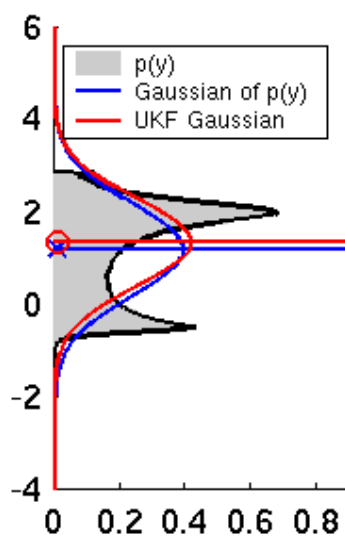
- **Highly efficient**: Polynomial in measurement dimensionality k and state dimensionality n :
 $O(k^{2.376} + n^2)$
- **Not optimal!** (why?)
- Can **diverge** if nonlinearities are large!
- Works surprisingly well even when all assumptions are mildly violated!



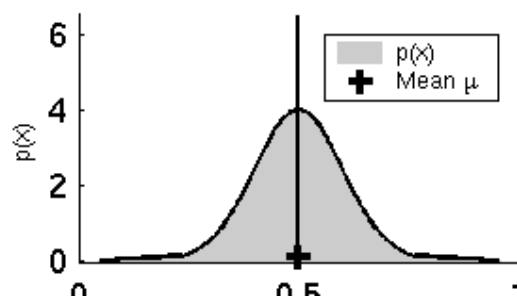
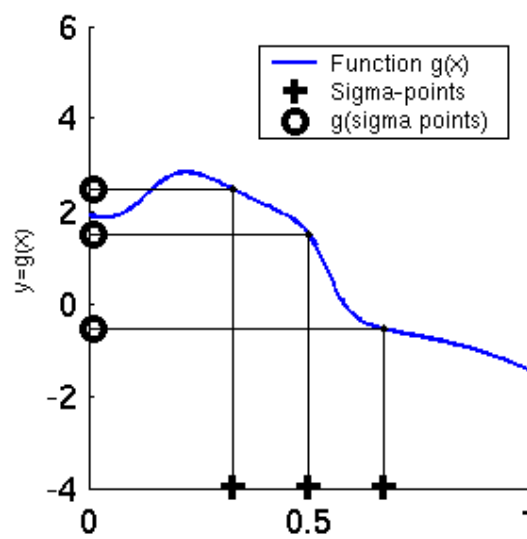
Linearization via Unscented Transform-1



EKF



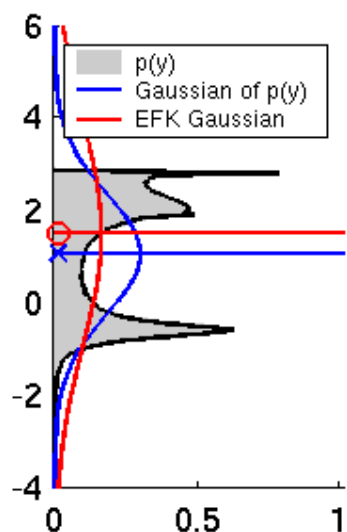
UKF



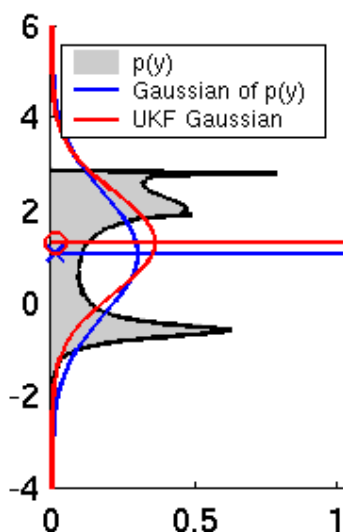
“Sigma-Points” calculated and passed through the nonlinear function. Pdf reconstructed from them.



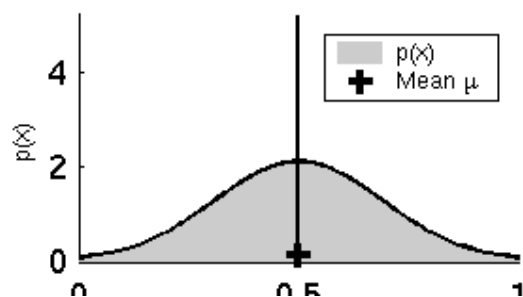
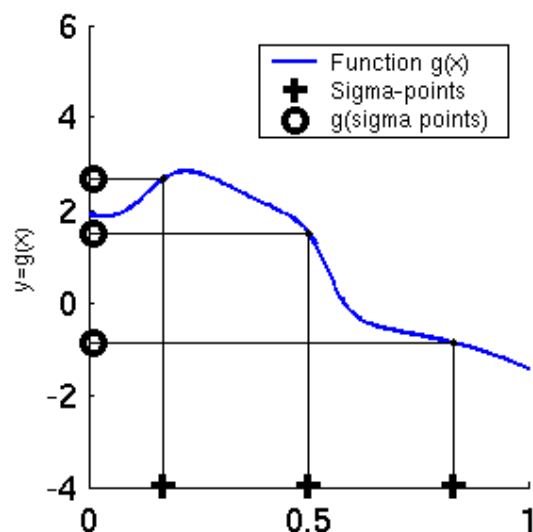
Linearization via Unscented Transform-2



EKF

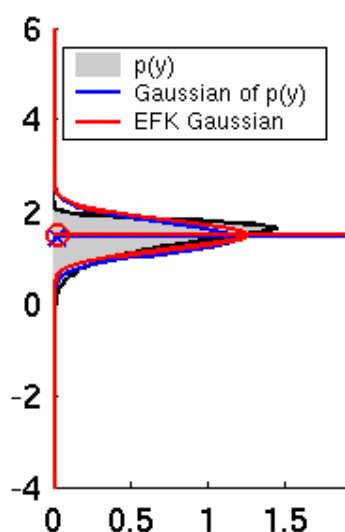


UKF

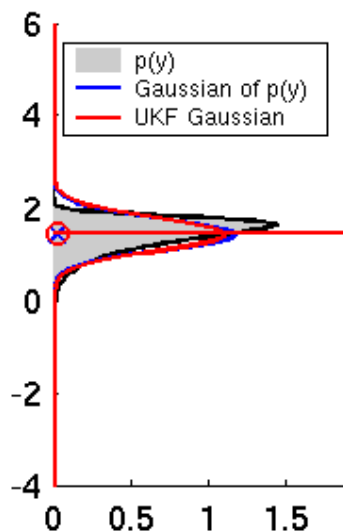




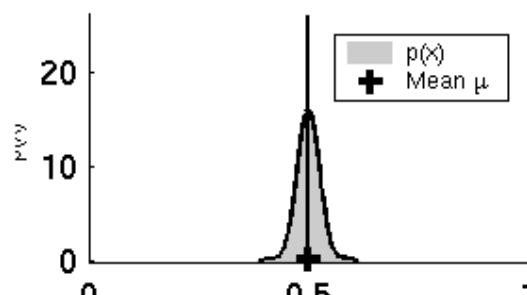
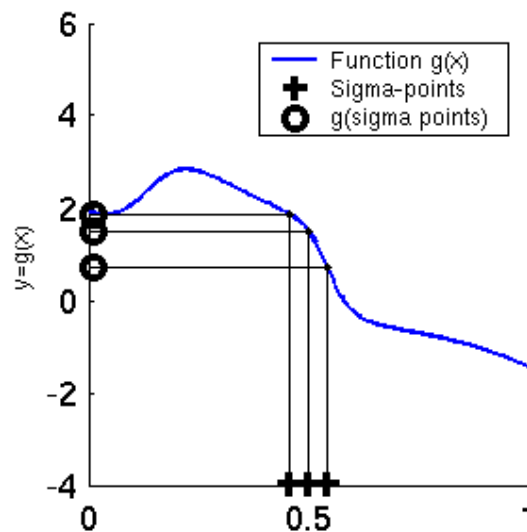
Linearization via Unscented Transform-3



EKF



UKF



**Better approximation performance under larger uncertainty than Taylor Series,
It result in a "derivative-free" filter!**



Unscented Transform

Find “Sigma points” and their “Pairs of Weights”

$$\begin{aligned}\chi^0 &= \mu & w_m^0 &= \frac{\lambda}{n + \lambda} & w_c^0 &= \frac{\lambda}{n + \lambda} + (1 - \alpha^2 + \beta) \\ \chi^i &= \mu \pm \left(\sqrt{(n + \lambda) \Sigma} \right)_i & w_m^i &= w_c^i = \frac{1}{2(n + \lambda)} & & \text{for } i = 1, \dots, 2n\end{aligned}$$

Pass sigma points through nonlinear function

$$\psi^i = g(\chi^i)$$

Recover mean and covariance

$$\begin{aligned}\mu' &= \sum_{i=0}^{2n} w_m^i \psi^i \\ \Sigma' &= \sum_{i=0}^{2n} w_c^i (\psi^i - \mu)(\psi^i - \mu)^T\end{aligned}$$



The Unscented Kalman Filter (UKF)

1: **Algorithm Unscented_Kalman_filter**($\mu_{t-1}, \Sigma_{t-1}, u_t, z_t$):

2: $\mathcal{X}_{t-1} = (\mu_{t-1} \quad \mu_{t-1} + \gamma\sqrt{\Sigma_{t-1}} \quad \mu_{t-1} - \gamma\sqrt{\Sigma_{t-1}})$

3: $\bar{\mathcal{X}}_t^* = g(u_t, \mathcal{X}_{t-1})$

4: $\bar{\mu}_t = \sum_{i=0}^{2n} w_m^{[i]} \bar{\mathcal{X}}_t^{*[i]}$

5: $\bar{\Sigma}_t = \sum_{i=0}^{2n} w_c^{[i]} (\bar{\mathcal{X}}_t^{*[i]} - \bar{\mu}_t)(\bar{\mathcal{X}}_t^{*[i]} - \bar{\mu}_t)^T + R_t$

6: $\bar{\mathcal{X}}_t = (\bar{\mu}_t \quad \bar{\mu}_t + \gamma\sqrt{\bar{\Sigma}_t} \quad \bar{\mu}_t - \gamma\sqrt{\bar{\Sigma}_t})$

7: $\bar{\mathcal{Z}}_t = h(\bar{\mathcal{X}}_t)$

8: $\hat{z}_t = \sum_{i=0}^{2n} w_m^{[i]} \bar{\mathcal{Z}}_t^{[i]}$

9: $S_t = \sum_{i=0}^{2n} w_c^{[i]} (\bar{\mathcal{Z}}_t^{[i]} - \hat{z}_t)(\bar{\mathcal{Z}}_t^{[i]} - \hat{z}_t)^T + Q_t$

10: $\bar{\Sigma}_t^{x,z} = \sum_{i=0}^{2n} w_c^{[i]} (\bar{\mathcal{X}}_t^{[i]} - \bar{\mu}_t)(\bar{\mathcal{Z}}_t^{[i]} - \hat{z}_t)^T$

11: $K_t = \bar{\Sigma}_t^{x,z} S_t^{-1}$

12: $\mu_t = \bar{\mu}_t + K_t(z_t - \hat{z}_t)$

13: $\Sigma_t = \bar{\Sigma}_t - K_t S_t K_t^T$

14: **return** μ_t, Σ_t

1: **Algorithm Kalman_filter**($\mu_{t-1}, \Sigma_{t-1}, u_t, z_t$):

2: $\bar{\mu}_t = A_t \mu_{t-1} + B_t u_t$

3: $\bar{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + R_t$

4: $K_t = \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1}$

5: $\mu_t = \bar{\mu}_t + K_t(z_t - C_t \bar{\mu}_t)$

6: $\Sigma_t = (I - K_t C_t) \bar{\Sigma}_t$

7: **return** μ_t, Σ_t



UKF Localization (1/2)

1: Algorithm UKF_localization($\mu_{t-1}, \Sigma_{t-1}, u_t, z_t, m$):

Generate augmented mean and covariance

$$2: M_t = \begin{pmatrix} \alpha_1 v_t^2 + \alpha_2 \omega_t^2 & 0 \\ 0 & \alpha_3 v_t^2 + \alpha_4 \omega_t^2 \end{pmatrix}$$

Motion noise

$$3: Q_t = \begin{pmatrix} \sigma_r^2 & 0 \\ 0 & \sigma_\phi^2 \end{pmatrix}$$

Measurement noise

$$4: \mu_{t-1}^a = (\mu_{t-1}^T \quad (0 \ 0)^T \quad (0 \ 0)^T)^T$$

Augmented state mean

$$5: \Sigma_{t-1}^a = \begin{pmatrix} \Sigma_{t-1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & M_t & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & Q_t \end{pmatrix}$$

Augmented covariance

Generate sigma points

$$6: \mathcal{X}_{t-1}^a = (\mu_{t-1}^a \quad \mu_{t-1}^a + \gamma \sqrt{\Sigma_{t-1}^a} \quad \mu_{t-1}^a - \gamma \sqrt{\Sigma_{t-1}^a})$$

2L+1 = 15 Sigma points
Each with components
for x, u, z

Pass sigma points through motion model and compute Gaussian statistics

$$7: \bar{\mathcal{X}}_t^x = g(u_t + \mathcal{X}_t^u, \mathcal{X}_{t-1}^x)$$

Prediction of sigma points

$$8: \bar{\mu}_t = \sum_{i=0}^{2L} w_i^{(m)} \bar{\mathcal{X}}_{i,t}^x$$

Predicted mean

$$9: \bar{\Sigma}_t = \sum_{i=0}^{2L} w_i^{(c)} (\bar{\mathcal{X}}_{i,t}^x - \bar{\mu}_t)(\bar{\mathcal{X}}_{i,t}^x - \bar{\mu}_t)^T$$

Predicted covariance



UKF Localization (2/2)

Predict observations at sigma points and compute Gaussian statistics

$$10: \quad \bar{\mathcal{Z}}_t = h(\bar{\mathcal{X}}_t^x) + \mathcal{X}_t^z$$

Measurement Sigma Points

$$11: \quad \hat{z}_t = \sum_{i=0}^{2L} w_i^{(m)} \bar{\mathcal{Z}}_{i,t}$$

Predicted measurement mean

$$12: \quad S_t = \sum_{i=0}^{2L} w_i^{(c)} (\bar{\mathcal{Z}}_{i,t} - \hat{z}_t)(\bar{\mathcal{Z}}_{i,t} - \hat{z}_t)^T$$

Predicted Meas. Covar.

$$13: \quad \Sigma_t^{x,z} = \sum_{i=0}^{2L} w_i^{(c)} (\bar{\mathcal{X}}_{i,t}^x - \bar{\mu}_t)(\bar{\mathcal{Z}}_{i,t} - \hat{z}_t)^T$$

Cross-Covariance

Update mean and covariance

$$14: \quad K_t = \Sigma_t^{x,z} S_t^{-1}$$

Kalman Gain

$$15: \quad \mu_t = \bar{\mu}_t + K_t(z_t - \hat{z}_t)$$

Updated Mean

$$16: \quad \Sigma_t = \bar{\Sigma}_t - K_t S_t K_t^T$$

Updated Covariance

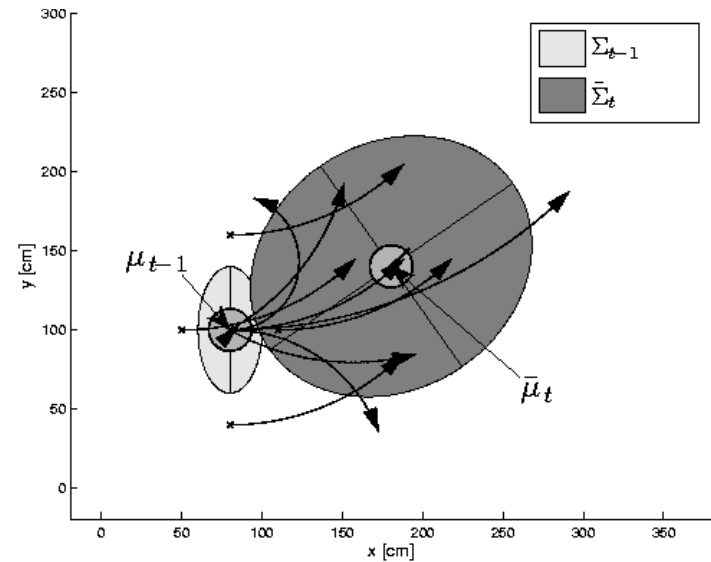
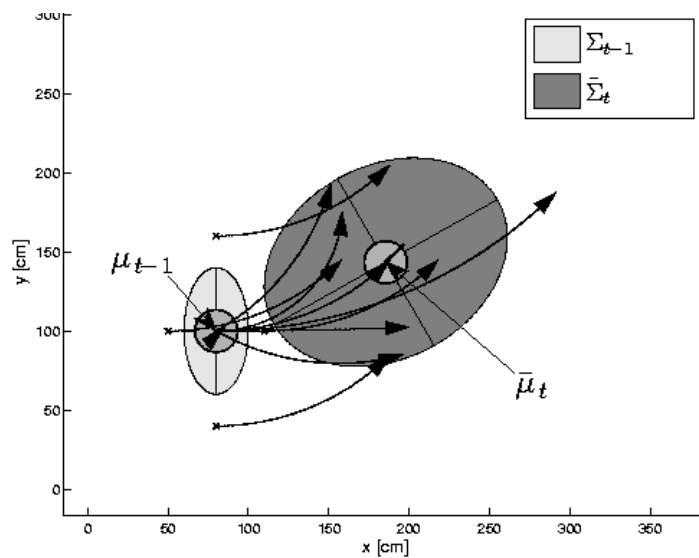
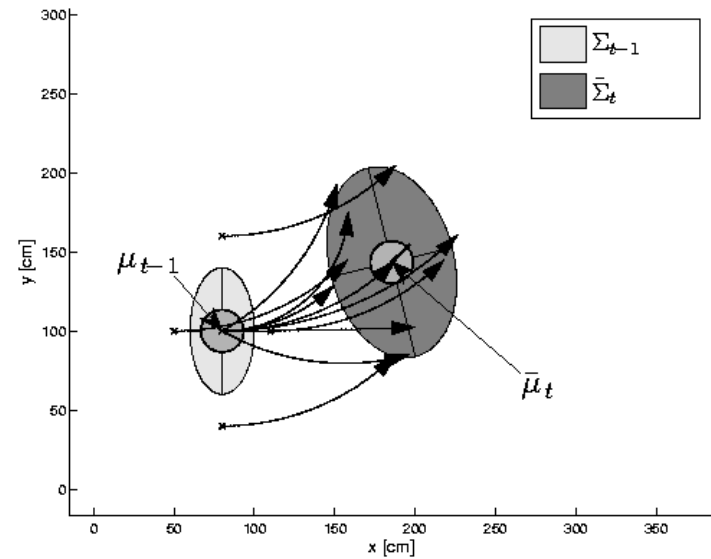
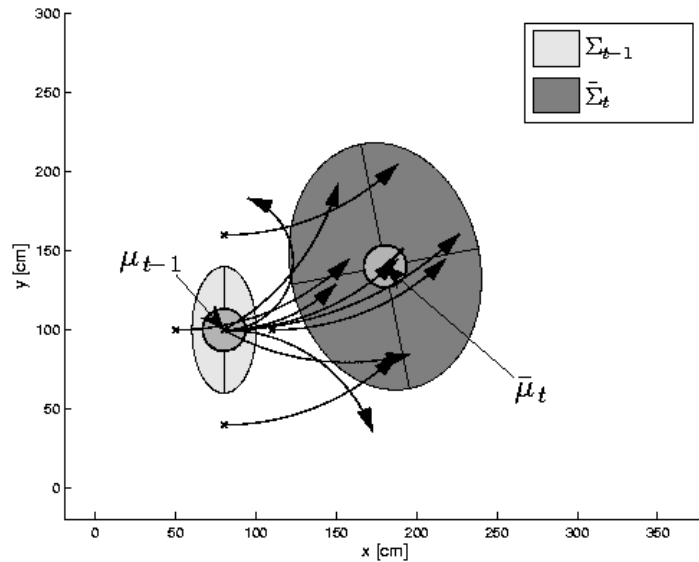
$$17: \quad p_{z_t} = \det(2\pi S_t)^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (z_t - \hat{z}_t)^T S_t^{-1} (z_t - \hat{z}_t) \right\}$$

$$18: \quad \text{return } \mu_t, \Sigma_t, p_{z_t}$$

Measurement Likelihood

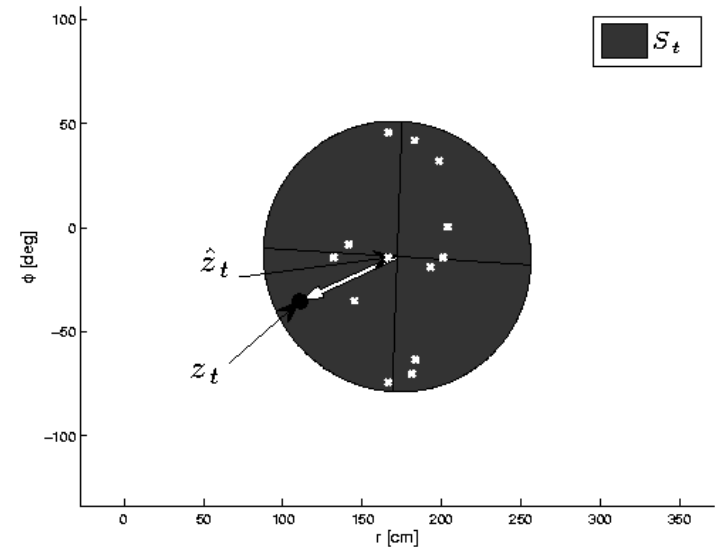
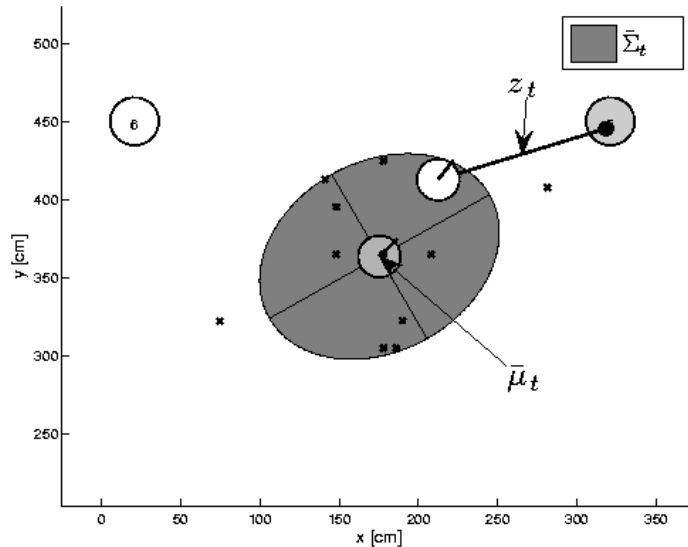
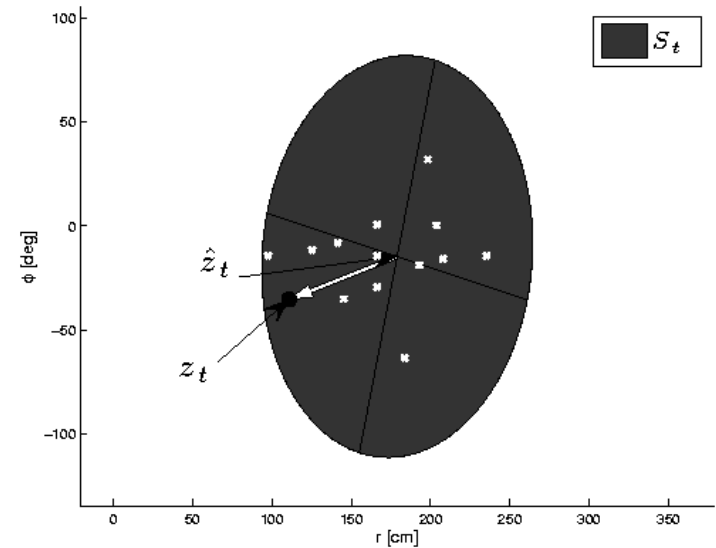
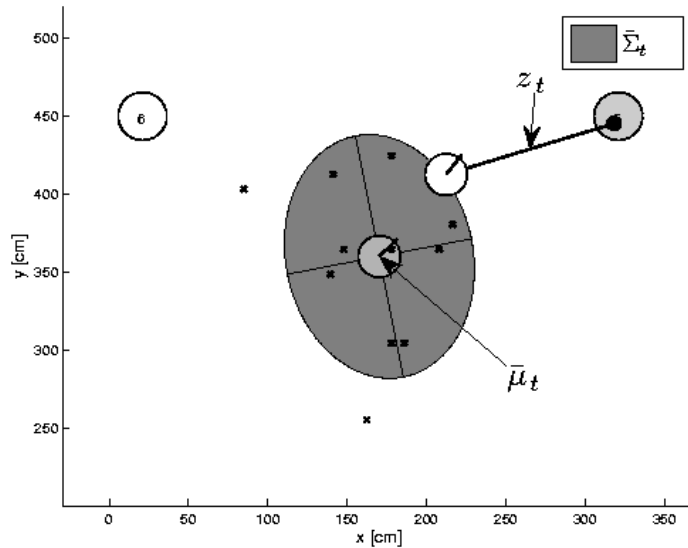


UKF Prediction Step



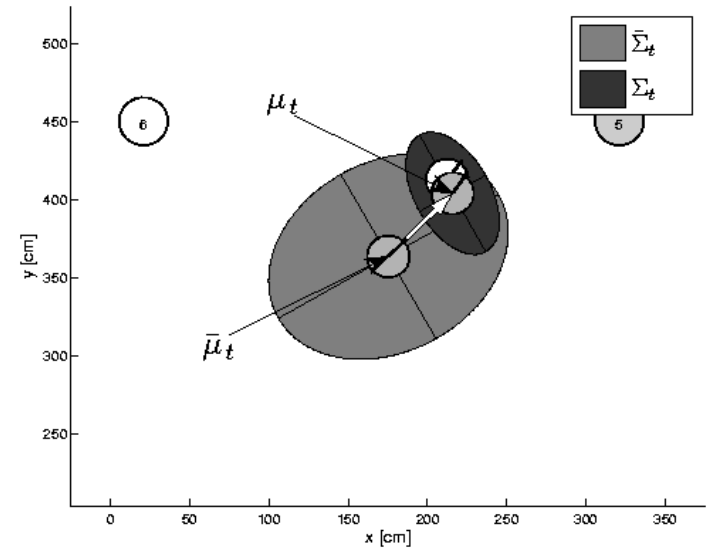
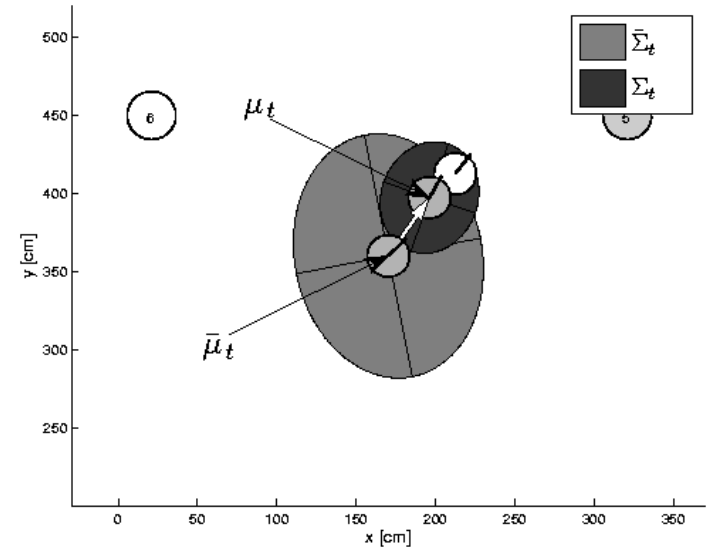
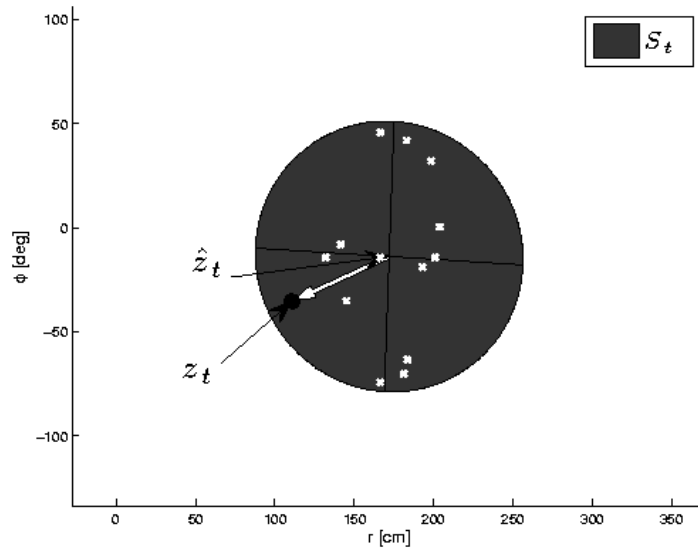
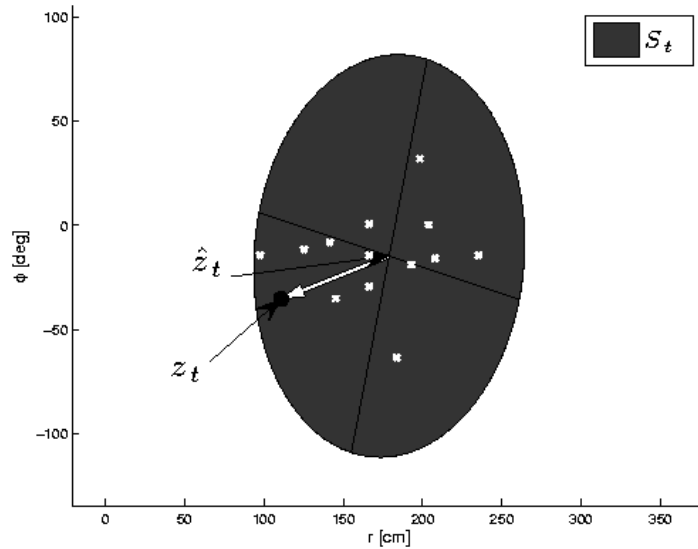


UKF Measurement Prediction Step



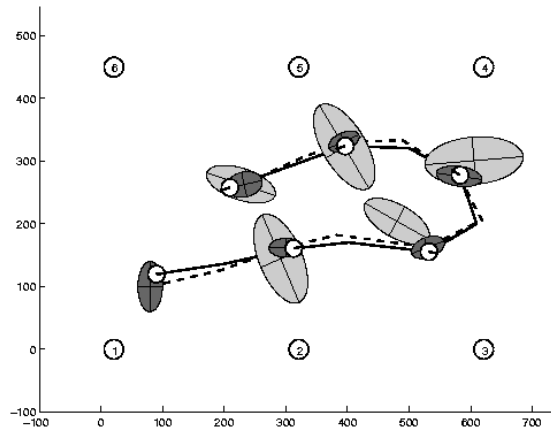


UKF Measurement Correction Step

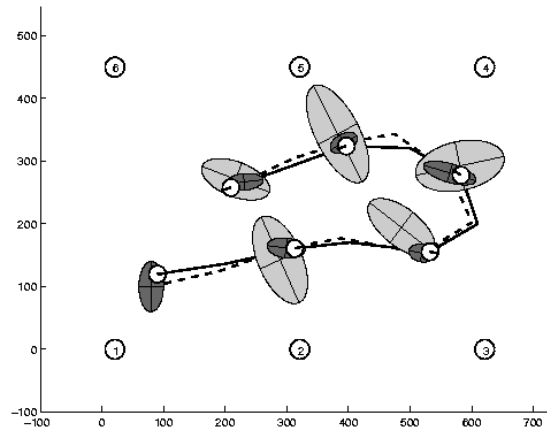




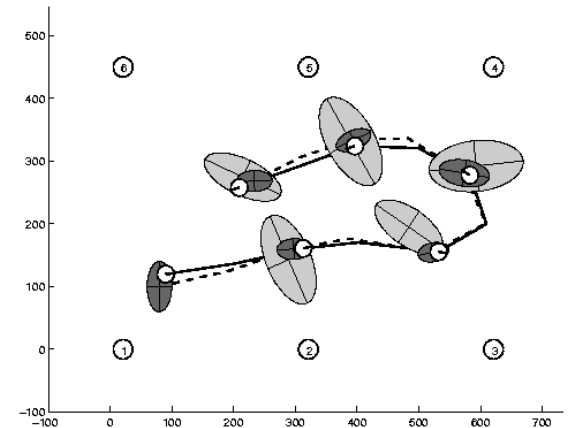
UKF Estimation Sequence Compared



EKF



PF

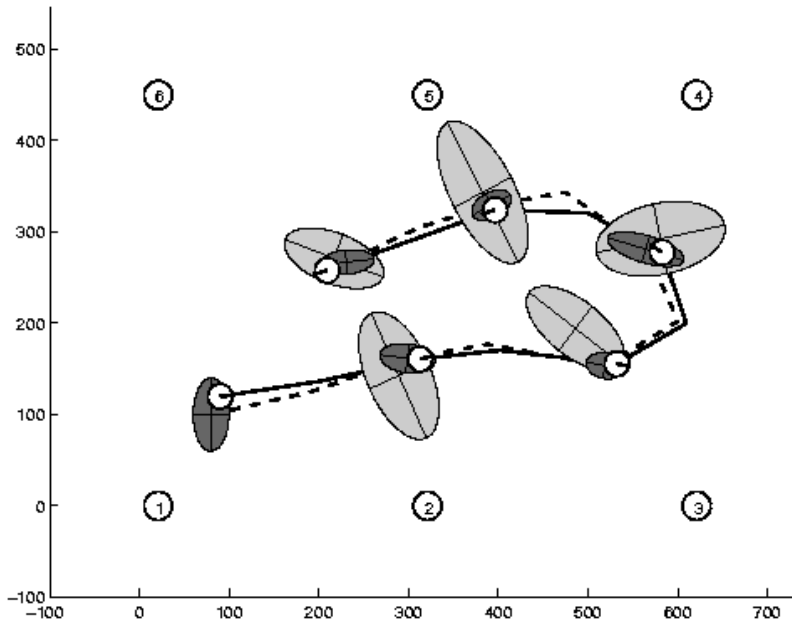


UKF

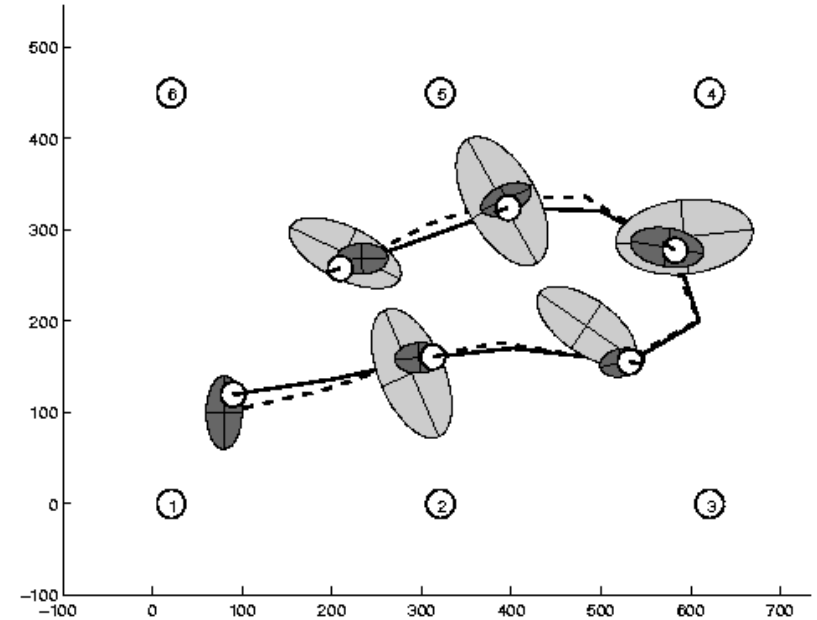
Both very close to reference Particle Filter in this example



UKF Estimation Sequence Compared



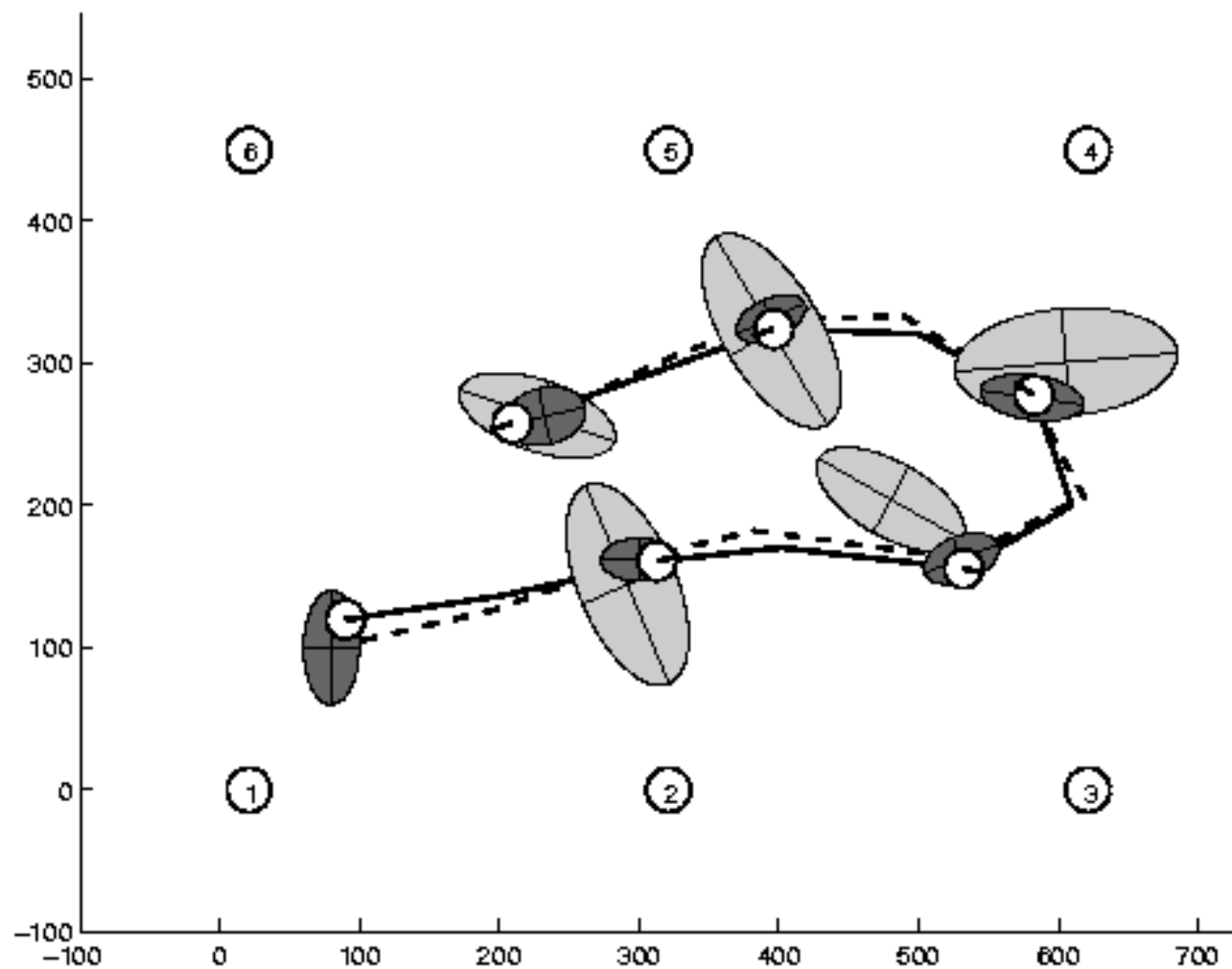
EKF



UKF

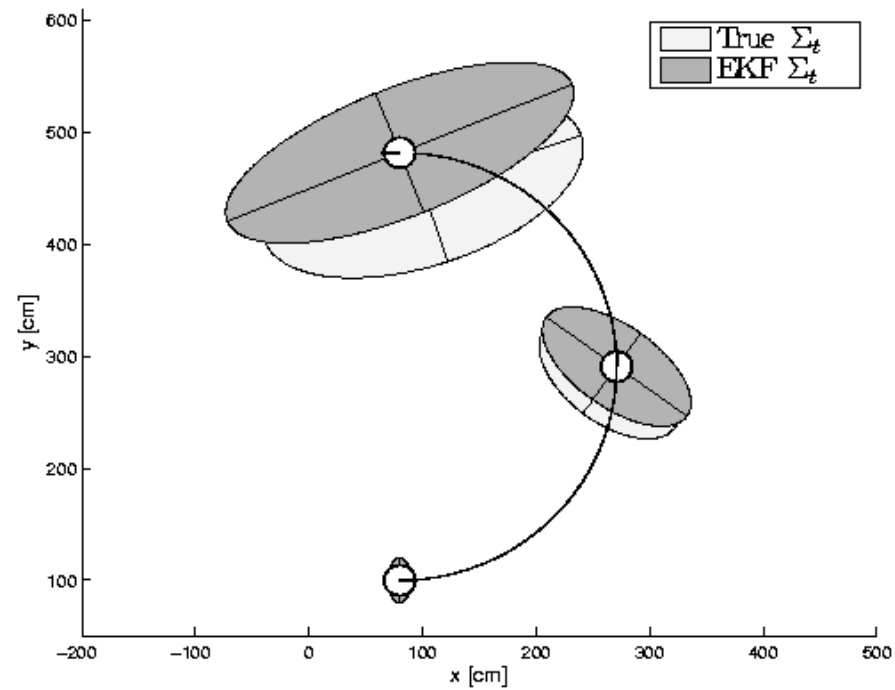


UKF Estimation Sequence Compared

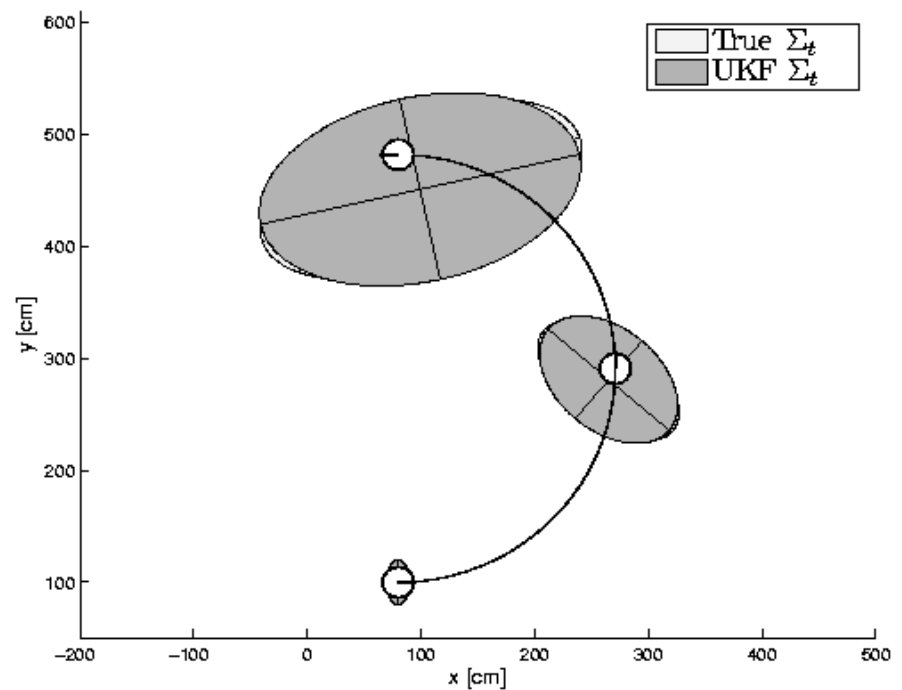




Prediction Quality



EKF



UKF



UKF Summary

- **Highly efficient:** Same complexity as EKF, with a constant factor slower in typical practical applications
- **Better linearization than EKF:**
Accurate in first two terms of Taylor expansion (EKF only first term)
- **Derivative-free:** No Jacobians needed
- **Still not optimal!**



Canonical Parametrization and the Information Filter

- We have exclusively used the “Moments Parametrization” of Gaussians: *Mean, Variance*,
- Dual Alternative: **“Canonical” Parametrization.**
(An Information Matrix + Information Vector)
- Results in “Information Filter” implementation,
- They have dual computational properties. Usually a difficult/complex step in one becomes an easy/light step in the other.
- They have nice properties. But much less explored and less popular as compared to KF, EKF, UKF.



Case of Unknown Correspondence

- What if we do not know the IDs of landmarks?
(Unknown correspondence of measurement with landmark ID)
- We have to deal with a “discrete uncertainty” besides the continuous uncertainty (state and measurement),
- *Most simple approach:*
Maximum Likelihood Correspondence
- ***“Pick the correspondence that maximizes the measurement likelihood”***
$$\hat{c}_t = \arg \max_{c_t} p(z_t \mid c_{1:t}, m, z_{1:t-1}, u_{1:t})$$
- The prediction step remains unchanged, the measurement update is modified:



Case of Unknown Correspondence

```

9:   for all observed features  $z_t^i = (r_t^i \ \phi_t^i \ s_t^i)^T$  do
10:     for all landmarks  $k$  in the map  $m$  do
11:        $q = (m_{k,x} - \bar{\mu}_{t,x})^2 + (m_{k,y} - \bar{\mu}_{t,y})^2$ 
12:        $\hat{z}_t^k = \begin{pmatrix} \sqrt{q} \\ \text{atan2}(m_{k,y} - \bar{\mu}_{t,y}, m_{k,x} - \bar{\mu}_{t,x}) - \bar{\mu}_{t,\theta} \\ m_{k,s} \end{pmatrix}$ 
13:        $H_t^k = \begin{pmatrix} -\frac{m_{k,x} - \bar{\mu}_{t,x}}{\sqrt{q}} & -\frac{m_{k,y} - \bar{\mu}_{t,y}}{\sqrt{q}} & 0 \\ \frac{m_{k,y} - \bar{\mu}_{t,y}}{q} & -\frac{m_{k,x} - \bar{\mu}_{t,x}}{q} & -1 \\ 0 & 0 & 0 \end{pmatrix}$ 
14:        $S_t^k = H_t^k \bar{\Sigma}_t [H_t^k]^T + Q_t$ 
15:     endfor
16:      $j(i) = \underset{k}{\operatorname{argmax}} \det(2\pi S_t^k)^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (z_t^i - \hat{z}_t^k)^T [S_t^k]^{-1} (z_t^i - \hat{z}_t^k) \right\}$ 
17:      $K_t^i = \bar{\Sigma}_t [H_t^{j(i)}]^T [S_t^{j(i)}]^{-1}$ 
18:      $\bar{\mu}_t = \bar{\mu}_t + K_t^i (z_t^i - \hat{z}_t^{j(i)})$ 
19:      $\bar{\Sigma}_t = (I - K_t^i H_t^{j(i)}) \bar{\Sigma}_t$ 
20:   endfor
21:    $\mu_t = \bar{\mu}_t$ 
22:    $\Sigma_t = \bar{\Sigma}_t$ 
23:   return  $\mu_t, \Sigma_t$ 

```

Landmark is selected based on
the measurement likelihood
from each landmark



Multi-Hypothesis Tracking (MHT)

- Belief is represented by multiple hypotheses
- Each hypothesis is tracked by a KF, EKF or UKF
- **Additional problems:**
 - **Data association:** Which observation corresponds to which hypothesis?
 - **Hypothesis management:** When to add / delete hypotheses?
- Huge body of literature on target tracking, motion correspondence etc.



Summary of Gaussian Filters

- Concrete implementations of the Bayes Filter when the dynamic system is Linear-Gaussian,
- Exact filtering algorithm depends on Gaussian Parametrization → *Kalman Filter (KF)* or *Information Filter (IF)*
- Nonlinearity is very common in applications. Depending on how approximate the nonlinearity by a linear system → *Extended Kalman Filter (EKF)* or *Unscented Kalman Filter (UKF)*
- UKF generally is a better approximation and is more robust against uncertainty.
- All filters are approximate when dealing with nonlinear system and/or non-gaussian noise.
- However, they are proven to be quite robust against mild violations of the assumptions and they are very efficient.



Summary of Markov(Gaussian) Localization - 1

- Simple application of the Gaussian filters to the Localization problem,
- Usually needs a “landmark based map”.
- In its simplest form, cannot deal with Global localization,
- But very efficient when Tracking Problems are considered,
- Still some issues in tracking problems with hard constraints on posterior (e.g., robot cannot be inside a wall),
- Filters usually works best if the uncertainty can be kept relatively small.
- MHT algorithm overcomes most of the above problems, with increased computational complexity,



Summary of Markov(Gaussian) Localization - 2

- **Global Localization can be solved with appropriate number of Gaussians in the Mixture,**
- **Kidnapped Robot Problem can be solved by injecting new Gaussians to the Mixture during filter operation,**
- **MHT better handles the problem of false correspondence (which can severely effect the single Gaussian EKF-UKF),**
- **MHT suffers from the same type of approximation errors as EKF and UKF since it uses the same linearization approximations**
- **Update equations for Gaussian Filters are quite complex mathematically but very efficient numerically!!**



Later On: SLAM Algorithms

- We will consider the family of algorithms that can perform “Simultaneous Localization and Mapping”
- State will be augmented to include map features as part of the “*state to be estimated*”,
- All computational structures will be modified to hold data related with these static portion of the state.
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